Chapter 4

Other Ways to Represent Solutions

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 4; printed pp. 159–236; PDF pp. 176–253.

Chinese reference: 中译本：第四章，书内第 111 页起；章首页 PDF 第 129 页。

Editorial study edition. This chapter reconstructs the source exposition, not its wording. Source theorem identifiers are stated separately from our numbering. Detailed proofs, Chinese mathematical annotations, exercise solutions and figures are editorial additions.

The organizing question is how a PDE can be represented by simpler objects:

$$\text{modes and profiles} \longrightarrow \text{integral transforms},$$
$$\text{changes of unknown} \longrightarrow \text{asymptotics and power series}.$$

Reading conventions. Fourier transforms use the unitary normalization $(2\pi)^{-n/2}$. The diffusion exponent is denoted by m to distinguish it from multiindices. A formal asymptotic calculation is identified explicitly; its conclusion is not promoted to an unconditional convergence theorem. We invoke standard integration, ODE theory, implicit and inverse function theorems, and elementary complex analysis. Spectral completeness and elliptic solvability are identified where used, rather than proved by writing a series. Sobolev terminology is available in Chapter 5.

表示公式还需要合法性

猜到一个形式不等于已经解决问题。接着要检查初边值、可逆性、级数或积分的收敛，以及能否逐项求导；渐近展开还需另估余项。

4.1 Separation of Variables

4.1.1 Separated modes and the missing convergence question

For $u_t - \Delta u = 0$ with zero Dirichlet data on a bounded domain U, try $u(x,t) = T(t)X(x)$. On a product neighborhood where both factors are nonzero,

$$\frac{T'}{T} = \frac{\Delta X}{X} = -\lambda.$$

Fixing x and then t shows that the common expression is constant. The separated equations are

$$-\Delta X = \lambda X, \quad X|_{\partial U} = 0, \qquad T' = -\lambda T.$$

The identities without division continue across zeros of X. For a nonzero smooth eigenfunction, integration by parts gives

$$\lambda \int_U |X|^2 = \int_U |DX|^2 > 0.$$

Strict positivity follows since a zero-gradient function with zero boundary trace is zero. Thus $e^{-\lambda t}X$ is a heat mode.

分离与叠加是两步

乘积形式只构造单个模态；把多个模态相加依赖方程的线性。非线性方程即使能分离变量，也通常不能把分离解相加来拟合任意初值。

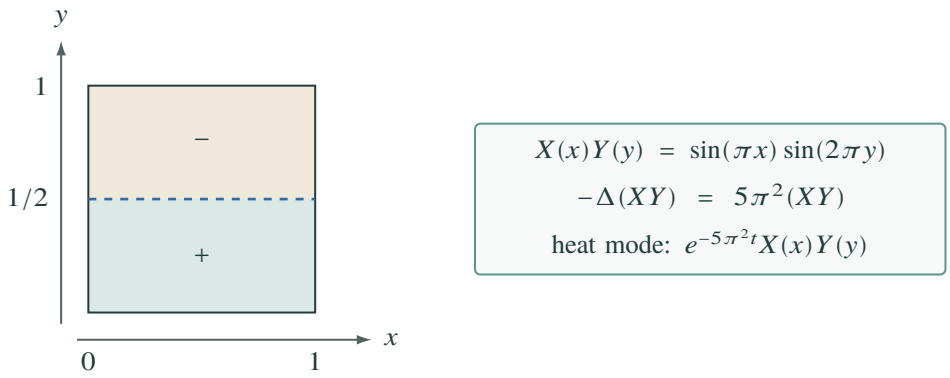


Diagram T4.1. A separated mode and its nodal line. On the unit square, this Dirichlet mode is the product of two one-dimensional modes. The dashed line $y = 1/2$ is a zero set inside the domain. Applying $-\Delta$ adds the two eigenvalues, giving $5\pi^2$; heat evolution then changes only the time factor.

Theorem 4.1 (A justified spectral representation, conditional on the spectral basis)

Assume the Dirichlet eigenfunctions X_j form an orthonormal basis of $L^2(U)$, with eigenvalues $\lambda_j > 0$. For $g \in L^2(U)$, set $g_j = (g, X_j)$ and

$$u(\cdot,t) = \sum_{j=1}^{\infty} e^{-\lambda_j t} g_j X_j. \tag{S4}$$

This converges in L^2, tends to g in L^2 at time zero, and solves the heat

> *equation in the spectral weak sense.*

Expanded proof.

Parseval gives $\sum |g_j|^2 = \|g\|_2^2$, so the tail norm squared is at most $\sum_{j>N} |g_j|^2$, uniformly for $t \geq 0$. Moreover

$$\|u(t) - g\|_2^2 = \sum_j |e^{-\lambda_j t} - 1|^2 |g_j|^2 \longrightarrow 0$$

by dominated convergence for a summable series. For $t \geq \tau > 0$, $\sup_{\lambda \geq 0} \lambda^k e^{-\tau\lambda} < \infty$. Hence time derivatives of every order converge in L^2 away from zero; the spectral Laplacian and time derivative agree termwise. Testing against any finite eigenfunction combination proves the weak equation, and density extends it to the corresponding energy test class. Interior classical smoothness requires the usual elliptic/local regularity facts; it does not follow simply from writing (S4). Existence and completeness of the basis are the later spectral theorem of Chapter 6, not a conclusion of separation itself. □

On $(0, \pi)$ one has $X_j = \sqrt{2/\pi}\sin(jx)$, $\lambda_j = j^2$. The coefficient formula is then an ordinary sine-series expansion. A finite sum is already a classical solution for its own initial datum.

4.1.2 Nonlinear and additive separation

For $u_t - \Delta(u^m) = 0$, $m > 1$, try $u = T(t)X(x) \geq 0$. Homogeneity gives $T'/T^m = \Delta(X^m)/X = \mu$ and therefore

$$T(t) = [c - (m-1)\mu t]^{-1/(m-1)}, \qquad \Delta(X^m) = \mu X.$$

Taking $X = |x|^\alpha$ and using $\Delta r^q = q(q+n-2)r^{q-2}$ for $r > 0$ yields

$$\alpha = \frac{2}{m-1}, \qquad \mu = \alpha m(\alpha m + n - 2) > 0.$$

The positive bracket restricts the solution interval to $t < c/((m-1)\mu)$. At the origin X need not be C^2, but $X^m = r^{2+2/(m-1)}$ is C^2 and its Laplacian extends continuously to zero. Thus the equation holds there in this natural sense. The profile grows at infinity and has infinite total mass.

For the Hamilton–Jacobi equation, additive separation is more natural: $u = X(x) + T(t)$ gives $H(DX) = \mu$ and $T = -\mu t + b$. The affine choice $X = a \cdot x$ recovers the complete family of Chapter 3.

4.1.3 Turing instability: diffusion can destabilize a coupled system

Let $J = D\mathbf{f}(0) = \left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right)$, and $D = \operatorname{diag}(d_1, d_2)$, $d_i > 0$. The reaction ODE is linearly asymptotically stable precisely when

$$\operatorname{tr} J = \alpha + \delta < 0, \qquad \det J = \alpha\delta - \beta\gamma > 0.$$

Indeed its two eigenvalues either are real with negative sum and positive product, or are conjugate with negative real part.

For $\mathbf{v}_t - D\Delta\mathbf{v} = J\mathbf{v}$, expand in Laplacian modes. The coefficient of X_j solves

$$s_j' = (J - \lambda_j D)s_j.$$

Its trace is still negative, but its determinant is

$$p(\lambda) = d_1 d_2 \lambda^2 - (d_1\delta + d_2\alpha)\lambda + \det J.$$

Theorem 4.2 (Linear mode instability)
A mode grows exponentially if and only if $p(\lambda_j) < 0$. For a continuous positive instability band it is necessary and sufficient that

$$B = d_1\delta + d_2\alpha > 0, \qquad B^2 > 4d_1 d_2 \det J.$$

Actual instability additionally requires a domain eigenvalue in (λ_-, λ_+), where

$$\lambda_\pm = \frac{B \pm \sqrt{B^2 - 4d_1 d_2 \det J}}{2d_1 d_2}.$$

Expanded proof.

Negative determinant gives two real eigenvalues of opposite signs. Positive determinant and negative trace give negative real parts; zero determinant is a neutral threshold, not strict exponential growth. The quadratic p opens upward and has positive roots exactly under the displayed conditions. Its negative set is the interval between them. The discrete spectrum of the chosen domain must meet this interval. □

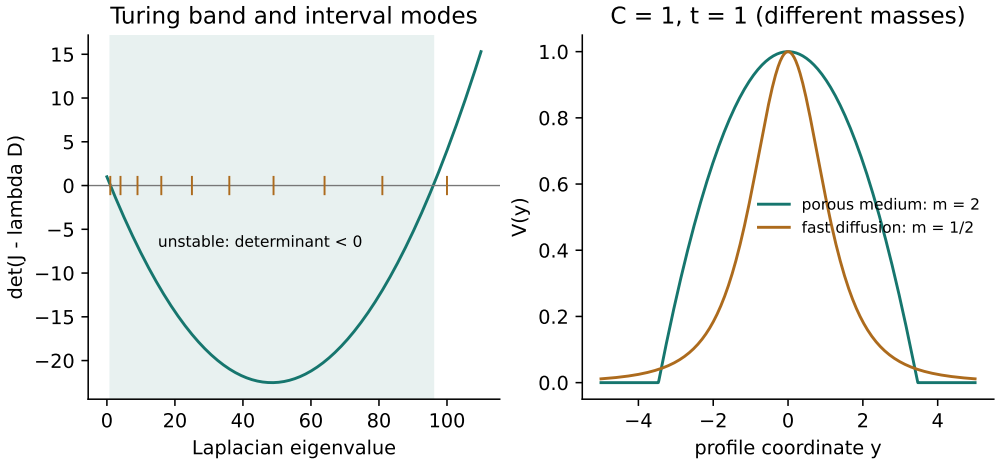
无穷远的条件影响物理解

这个分离解能在有限时间爆破，但它从一开始就有无限质量。不能用它否定有限质量、紧支集数据下的扩散结论；全空间问题也要记录增长类。

For example, take

$$J = \begin{pmatrix} 1 & -2 \\ 2 & -3 \end{pmatrix}, \qquad D = \operatorname{diag}(0.01, 1).$$

Then $\operatorname{tr} J = -2$, $\det J = 1$, while $p(4) = 0.16 - 3.88 + 1 = -2.72$. On $(0, \pi)$ the second sine mode has $\lambda_2 = 4$ and grows. If $D = dI$, by contrast, every eigenvalue is shifted by $-d\lambda_j$, so a stable J remains stable.



Editorial figure. The determinant band is compared with the actual interval spectrum; the other panel compares scaling profiles.

4.2 Similarity Solutions

4.2.1 Plane waves, phase velocity and group velocity

For $u = e^{i(\xi\cdot x - \omega t)}$, direct substitution gives:

equation	dispersion relation
$u_t - \Delta u = 0$	$\omega = -i\lvert\xi\rvert^2$
$u_{tt} - \Delta u = 0$	$\omega = \pm\lvert\xi\rvert$
$u_{tt} - \Delta u + M^2 u = 0$	$\omega = \pm\sqrt{\lvert\xi\rvert^2 + M^2}$
$iu_t + \Delta u = 0$	$\omega = \lvert\xi\rvert^2$
$u_t + u_{xxx} = 0$	$\omega = -\xi^3$

For instance $\partial_{xxx} e^{i\xi x} = -i\xi^3 e^{i\xi x}$, so the last sign is negative with this phase convention. A real ω gives oscillation; a negative imaginary part gives damping. For $\xi \neq 0$, the signed phase speed in direction $\xi/|\xi|$ is $\omega/|\xi|$.

To explain group velocity, set $x = tv$ in a Fourier superposition:

$$u(tv, t) = \int e^{it(\xi\cdot v - \omega(\xi))} a(\xi)\, d\xi.$$

连续波段不等于域内存在模态

代数条件先给出不稳定的 λ 区间；还要看具体区域的离散特征值是否落入其中。$p = 0$ 只是临界中性情形，严格增长需要 $p < 0$。

线性失稳不等于已证明斑图形成

这里证明的是线性化系统存在增长模态。非线性解是否饱和、形成何种稳定空间图案，需要额外的分岔与演化理论，不能由特征值计算直接得出。

Stationary points satisfy $v = D_\xi \omega$. The stationary-phase theorem below makes this observation precise for localized amplitudes and nondegenerate stationary points.

4.2.2 KdV solitary waves

For $u_t + 6uu_x + u_{xxx} = 0$, take $u = v(s)$, $s = x - ct$. The ODE is $-cv' + 6vv' + v''' = 0$.

Theorem 4.3 (The source solitary-wave calculation)
For $c > 0$ and $s_0 \in \mathbb{R}$,

$$u(x,t) = \frac{c}{2}\operatorname{sech}^2\left(\frac{\sqrt{c}}{2}(x - ct - s_0)\right)$$

is a decaying traveling wave.

Expanded proof.

One integration gives $v'' - cv + 3v^2 = A$. Multiply by v' and integrate again:

$$\frac{1}{2}(v')^2 = \frac{c}{2}v^2 - v^3 + Av + B.$$

The solitary-wave limits $v, v', v'' \to 0$ at both ends force $A = B = 0$. A positive hump therefore has maximum $c/2$ and obeys $v' = -v\sqrt{c - 2v}$ on its decreasing half. With $v = (c/2)\operatorname{sech}^2\theta$,

$$\frac{dv}{v\sqrt{c - 2v}} = -\frac{2}{\sqrt{c}}\,d\theta \quad (\theta > 0).$$

Thus $\theta = \sqrt{c}(s - s_0)/2$ on that half. Reflection supplies the other half. For a direct check use $(\operatorname{sech}^2 z)'' = 4\operatorname{sech}^2 z - 6\operatorname{sech}^4 z$, which gives $v'' = cv - 3v^2$ everywhere. Differentiating verifies the original PDE, including the crest. □

A solitary-wave formula does not by itself prove the stronger scattering properties often associated with the word soliton. Also the KdV sign in Problems 11–12 is the opposite nonlinear sign; there $u \mapsto -u$ relates the conventions.

4.2.3 Bistable fronts: a detailed shooting argument

Assume $f(0) = f(a) = f(1) = 0$, $0 < a < 1$, $f < 0$ on $(0,a)$, $f > 0$ on $(a,1)$, $f'(0), f'(1) < 0$, and $\int_0^1 f > 0$. An increasing front from zero to one satisfies

$$v' = w, \qquad w' = -cw - f(v),$$
$$(v,w)(-\infty) = (0,0), \quad (v,w)(+\infty) = (1,0).$$

相速不是信息传播速度

Klein–Gordon 的正频相速为 $\sqrt{|\xi|^2 + M^2}/|\xi| > 1$，群速度的大小却小于一。单一相位面的移动不能直接当作信号速度或支集传播速度。

不能把减号分支用在整个波峰两侧

一阶能量式含正负两个分支。上升半支与下降半支要在 $v = c/2$ 处拼合；最后直接核验二阶 ODE，才能排除分支选择造成的假解。

At each endpoint the linear eigenvalues are $(-c \pm \sqrt{c^2 - 4f'(j)})/2, j = 0, 1$. They have opposite signs. We use the standard local stable/unstable manifold theorem for these hyperbolic ODE equilibria, including continuous dependence on c. The following fills in the source's phase-plane shooting outline.

Put $F(v) = \int_0^v f$ and $E = w^2/2 + F(v)$. Then $dE/ds = -cw^2$. Since $F(1) > 0$, a front must have $c < 0$:

$$c \int_{\mathbb{R}} (v')^2 \, ds = -\int_0^1 f(v) \, dv. \tag{BF4}$$

Choose a fixed $b \in (a, 1)$ so close to a that $F(b) < 0$. For $c = 0$ the upper unstable branch from zero and upper stable branch into one meet the section $v = b$ at heights

$$w_L(b; 0) = \sqrt{-2F(b)}, \qquad w_R(b; 0) = \sqrt{2(F(1) - F(b))}.$$

The first is smaller.

Expanded proof (existence and uniqueness of the front speed).

Where $w > 0$, write each trajectory as a graph in v:

$$\frac{dw}{dv} = -c - \frac{f(v)}{w}. \tag{SH4}$$

The endpoint slopes are $w_L/v \to \lambda_0^+(c)$ and $w_R/(1 - v) \to -\lambda_1^-(c)$. As c decreases, the first slope strictly increases and the second strictly decreases. At a first contact between two left branches, their slope difference from (SH4) is the positive difference of $-c$; this prevents crossing. Integrating from the right reverses the comparison for right branches. Thus $w_L(b; c)$ increases strictly as c decreases, while $w_R(b; c)$ decreases strictly.

For $c \le 0$ the left branch stays above its $c = 0$ graph up to b, so cannot stop by hitting zero. The right branch also reaches b: on $(b, 1), f > 0$, and at a hypothetical earlier endpoint with $w = 0$ the equation $w_s = -f < 0$ would give negative w immediately to its right in v, contradicting the positive branch. There is no finite-v upward blowup in these intervals since (SH4) has bounded positive slope whenever w is large. Along the right branch $E(v) \le F(1)$, since energy is increasing in forward time; hence $w_R(b; c) \le \sqrt{2(F(1) - F(b))}$.

To force the left height arbitrarily large, fix $K > 0$. The quantity $|f(v)/v|$ is bounded on $(0, b]$, say by C. On the line $w = Kv$,

$$\frac{dw}{dv} = -c - \frac{f(v)}{Kv} \ge -c - C/K > K$$

when $-c$ is sufficiently large, and $\lambda_0^+(c) > K$ then. The left branch starts above this line and cannot cross downward; so $w_L(b;c) \geq Kb$. The two heights therefore exchange order for sufficiently negative c. Continuous dependence and the intermediate value theorem give a matching speed; strict comparison makes it unique. At the match, ODE uniqueness joins the two trajectories. The hyperbolic endpoint asymptotics take infinite s to reach the endpoints, giving a front on all $\mathbb{R}$, unique up to translation. Integration of the energy identity proves (BF4). □

速度符号取决于连接方向

本稿取 $v(-\infty) = 0$、$v(+\infty) = 1$，故 $\int_0^1 f > 0$ 推出速度为负。若把波形左右反转，速度也反号，不能脱离行波坐标与端态单记正负。

4.2.4 Scaling and the Barenblatt solution

For $u_t - \Delta(u^m) = 0$, try $u = t^{-\alpha}V(y)$, $y = xt^{-\beta}$. The chain rule gives

$$u_t = t^{-\alpha-1}(-\alpha V - \beta y \cdot DV), \qquad \Delta u^m = t^{-\alpha m-2\beta}\Delta_y V^m.$$

Balancing time powers requires $\alpha + 1 = \alpha m + 2\beta$. Finite nonzero time-independent mass requires $\alpha = n\beta$, since $\int u = t^{-\alpha+n\beta}\int V$. Consequently

$$\beta = \frac{1}{n(m-1)+2}, \qquad \alpha = n\beta.$$

For a radial profile $V = W(r)$, the equation integrates to

$$[r^{n-1}(W^m)' + \beta r^n W]' = 0.$$

Regularity and zero flux at $r = 0$ set the integration constant to zero. For $m > 1$, on the positive set,

$$(W^{m-1})' = -\frac{m-1}{m}\beta r, \quad V(y) = (C - k|y|^2)_+^{1/(m-1)}, \quad k = \frac{(m-1)\beta}{2m}.$$

Theorem 4.4 (Barenblatt profile, source §4.2.2)
For $C > 0$, $m > 1$, the above formula defines a nonnegative distributional solution on $\mathbb{R}^n \times (0,\infty)$, with fixed finite mass. Its support radius is $\sqrt{C/k}\,t^\beta$.

Expanded proof (including the free-boundary check).

Inside its positive set the preceding identities prove the PDE. At the moving interface u tends to zero, and so the time derivative of the zero extension has no jump-mass term. Furthermore $V^m = (C - kr^2)_+^{m/(m-1)}$ has zero normal derivative at the interface because $m/(m-1) > 1$. Thus the distributional Laplacian adds no surface delta. The interior time derivative and second spatial derivatives are locally integrable up to the interface: the worst boundary power is $d^{1/(m-1)-1}$, whose exponent is greater than -1. Integration by parts on interior truncated regions, followed by

this integrability and the zero traces, proves the weak PDE across the boundary. Mass conservation follows by the change $x = t^{\beta} y$. Finally for compactly supported continuous ϕ,

$$\int u(x,t)\phi(x)\,dx = \int V(y)\phi(t^{\beta}y)\,dy \longrightarrow \phi(0)\int V.$$

The initial datum is a point mass, not a fixed bounded function. □

The radius grows sublinearly for $m > 1$. Its derivative is finite for each $t > 0$ but diverges as $t \downarrow 0$; finite propagation here means compact support at each positive time, not a uniform speed bound down to the point-source initial time. The fast-diffusion counterpart in Problem 5 has algebraic tails.

自由边界处要检查通量

只在正集内代入公式还不够。这里 $u = 0$ 消去移动边界的时间接口项，$\partial_\nu(u^m) = 0$ 消去空间接口项；扩展后的 PDE 才在分布意义下成立。

4.3 Transform Methods

4.3.1 Fourier transform: normalization and Plancherel

For $f \in L^1(\mathbb{R}^n)$ define

$$\hat{f}(\xi) = (2\pi)^{-n/2}\int e^{-ix\cdot\xi} f(x)\,dx,$$
$$\mathscr{F}^{-1}h(x) = (2\pi)^{-n/2}\int e^{ix\cdot\xi} h(\xi)\,d\xi.$$

The integrals are absolutely convergent. For Schwartz functions integration by parts and Fubini yield

$$\widehat{\partial_j f} = i\xi_j \hat{f}, \qquad \widehat{f * h} = (2\pi)^{n/2}\hat{f}\,\hat{h}. \tag{F4}$$

For the second identity, set $x = z + y$ in the double integral. Both normalizations must be retained; $\mathscr{F}$ is unitary, whereas ordinary convolution has no normalizing factor.

卷积公式为何多一个常数

每个 Fourier 变换自带 $(2\pi)^{-n/2}$，两个变换相乘就出现两份。卷积只做一次变换，所以要乘回 $(2\pi)^{n/2}$；后面的核常数都由此决定。

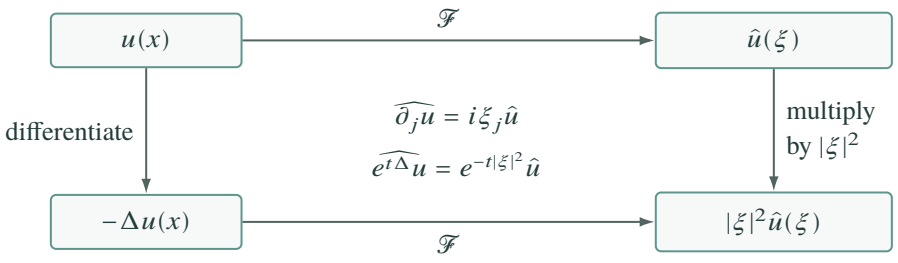


Diagram T4.2. Fourier transformation turns differentiation into multiplication. The square commutes with the unitary normalization used here. There is no 2π in the multiplier because the exponential is $e^{-ix\cdot\xi}$. The heat multiplier is small at large $|\xi|$, connecting this representation with the stability picture in Chapter 1.

Theorem 4.5 (Plancherel and inversion; source Theorems 1–2)
The Fourier transform extends uniquely to a unitary operator on L^2. It preserves inner products, has the displayed inverse, and $\widehat{D^\gamma f} = (i\xi)^\gamma \hat{f}$ whenever the weak derivative is in L^2.

Expanded proof (Gaussian regularization without circular inversion).

1. The Gaussian transform. For $a > 0$, $I(b) = \int_{\mathbb{R}} e^{-ax^2} e^{-ibx}\, dx$. Differentiate under the integral and integrate by parts to get $I'(b) = -bI(b)/(2a)$. Since $I(0) = \sqrt{\pi/a}$, $I(b) = \sqrt{\pi/a}\, e^{-b^2/(4a)}$. Products give the n-dimensional identity. This real-variable proof does not yet require contour deformation.

2. The norm identity. For $f \in L^1 \cap L^2$ and $\eta > 0$, Fubini gives

$$\int e^{-\eta|\xi|^2} |\hat{f}(\xi)|^2\, d\xi = \iint f(x)\overline{f(y)} G_\eta(x-y)\, dx\, dy,$$

where $G_\eta(z) = (4\pi\eta)^{-n/2} e^{-|z|^2/(4\eta)}$. Fubini is legitimate since $\|f\|_1^2 < \infty$ and the frequency Gaussian is integrable. The right side tends to $\|f\|_2^2$ because $G_\eta * f \to f$ in L^2. For completeness,

$$\|G_\eta * f - f\|_2 \le \int G_\eta(z) \|f(\cdot - z) - f\|_2\, dz \longrightarrow 0 :$$

translations are continuous in L^2 by approximation with compact continuous functions; split the last integral into a small ball and its Gaussian tail. Monotone convergence on the left gives the norm identity. Polarization gives the inner-product identity.

3. Inversion and extension. For Schwartz f, (F4) implies rapid decay of $\hat{f}$. Fubini and the Gaussian formula give

$$\mathscr{F}^{-1}(e^{-\eta|\xi|^2}\hat{f}) = G_\eta * f.$$

The right side tends to f; the left tends to $\mathscr{F}^{-1}\hat{f}$ by dominated convergence. Thus inversion holds on Schwartz functions. Approximate arbitrary L^2 data by Schwartz functions. The norm identity makes their transforms Cauchy and makes the limit independent of the approximation. Inversion extends by continuity, so the range is all of L^2. Finally integration by parts against Schwartz tests proves the derivative identity in distributions; if both sides belong to L^2, it is an a.e. equality of those representatives. □

4.3.2 Fourier multipliers and evolution equations

Bessel potential. For $-\Delta u + u = f, f \in L^2$, $\hat{u} = (1 + |\xi|^2)^{-1}\hat{f}$. This gives a unique L^2 distributional solution with $\int (1 + |\xi|^2)^2 |\hat{u}|^2 < \infty$. A rigorous kernel representation is

$$K(x) = \int_0^\infty e^{-s} G_s(x)\, ds, \qquad u = K * f.$$

Tonelli gives $K \geq 0$ and $\|K\|_1 = 1$; its transform is $(2\pi)^{-n/2}/(1 + |\xi|^2)$. The kernel called B in the source is $(2\pi)^{n/2}K$. The singularity of K at zero in some dimensions does not prevent its L^1 integrability or convolution action on L^2.

Heat and Schrödinger equations. Spatial transformation of $u_t - \Delta u = 0$ gives $\hat{u}_t + |\xi|^2 \hat{u} = 0$, hence $\hat{u} = e^{-t|\xi|^2}\hat{g}$. Gaussian inversion recovers $u = G_t * g$ from Chapter 2. For $iu_t + \Delta u = 0$ the sign is instead $\hat{u}_t = -i|\xi|^2\hat{u}$, so

$$\hat{u} = e^{-it|\xi|^2}\hat{g}, \qquad u(x,t) = (4\pi i t)^{-n/2} \int e^{i|x-y|^2/(4t)} g(y)\, dy. \tag{SC4}$$

For $t \neq 0$ use $(it)^{-n/2} = |t|^{-n/2} e^{-i\pi n\, \mathrm{sgn}(t)/4}$. The kernel identity follows by first replacing it with $\eta + it$, $\eta > 0$, and then letting $\eta \downarrow 0$. For $g \in L^1 \cap L^2$, the kernels converge under the spatial integral by domination, while the multipliers converge in L^2. Thus both descriptions agree. For L^2 data this defines a unitary group:

$$\|u(t)\|_2 = \|g\|_2, \qquad \|u(t) - g\|_2^2 = \int |e^{-it|\xi|^2} - 1|^2 |\hat{g}|^2 \longrightarrow 0.$$

If $g, |x|^2 g \in L^1$, differentiation of (SC4) on compact space-time sets away from $t = 0$ verifies the classical PDE. For arbitrary L^2 data the statement is distributional.

Wave and telegraph equations. For $u_{tt} - \Delta u = 0, u(0) = g, u_t(0) = h$,

$$\hat{u}(\xi, t) = \cos(t|\xi|)\hat{g} + \frac{\sin(t|\xi|)}{|\xi|}\hat{h}. \tag{WV4}$$

At $\xi = 0$ the quotient is defined as t. It is bounded by t, so the formula has no genuine zero-frequency pole. For $u_{tt} + 2du_t - u_{xx} = 0$ use $\hat{u} = e^{-dt}v$. Then $v'' + (\xi^2 - d^2)v = 0$ and

$$\hat{u} = e^{-dt} \begin{cases} \cosh(qt)\hat{g} + \dfrac{\sinh(qt)}{q}(\hat{h} + d\hat{g}), & |\xi| < d,\ q = \sqrt{d^2 - \xi^2}, \\ \hat{g} + t(\hat{h} + d\hat{g}), & |\xi| = d, \\ \cos(qt)\hat{g} + \dfrac{\sin(qt)}{q}(\hat{h} + d\hat{g}), & |\xi| > d,\ q = \sqrt{\xi^2 - d^2}. \end{cases}$$

L^2 变换不必是逐点积分

一般 L^2 函数未必可积，其 Fourier 变换通过范数极限定义。反演和乘子公式首先是 L^2 等式；若要逐点表示，还须增加可积性或正则性。

乘子有界不表示核处处有限

$(1 + |\xi|^2)^{-1}$ 给出有界的 L^2 乘子。对应核可能在原点无界，但仍可积；不能把一个点的奇性误当作表示公式失效。

色散与耗散的范数表现不同

热乘子模长小于一，会压低高频；Schrödinger 乘子模长为一，保持 L^2。后者的点态衰减来自振荡抵消，不来自 L^2 质量损失。

These are the overdamped, critical and oscillatory regimes. The common limiting formula at $q = 0$ is essential.

Theorem 4.6 (Asymptotic energy equipartition; source §4.3.1)
If $g \in H^1(\mathbb{R}^n)$ and $h \in L^2(\mathbb{R}^n)$, the wave solution has

$$\|Du(t)\|_2^2 \longrightarrow E_0, \qquad \|u_t(t)\|_2^2 \longrightarrow E_0, \qquad E_0 = \tfrac{1}{2}(\|Dg\|_2^2 + \|h\|_2^2).$$

Expanded proof.

Write $A = |\xi|\hat{g}$ and $B = \hat{h}$ in L^2. Plancherel and (WV4) give

$$\|Du\|_2^2 = \int |A|^2 \cos^2(t|\xi|) + |B|^2 \sin^2(t|\xi|) + 2\,\mathrm{Re}(A\overline{B}) \sin(t|\xi|)\cos(t|\xi|).$$

All three coefficient functions are L^1, the cross term by Cauchy–Schwarz. For any $F \in L^1(\mathbb{R}^n)$, polar coordinates give $\int e^{2it|\xi|}F(\xi)\,d\xi = \int_0^\infty e^{2itr}H(r)\,dr$ with $\|H\|_1 \le \|F\|_1$. The latter tends to zero: approximate H in L^1 by smooth functions supported inside $(0,\infty)$ and integrate those by parts. Now use the double-angle identities to obtain the first limit. The analogous formula for u_t gives energy conservation and the second limit. This also handles $n = 1$ without discarding a possible endpoint term in a radial integration by parts. □

4.3.3 Radon transform and plane-wave decomposition

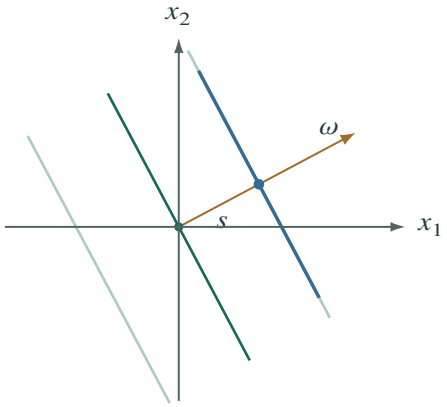
For $f \in C_c^\infty(\mathbb{R}^n)$, set

$$Rf(s,\omega) = \int_{x\cdot\omega = s} f(x)\,dS_x, \qquad \omega \in S^{n-1}.$$

Coordinates $x = s\omega + \sum y_j b_j$, where b_j are an orthonormal basis of $\omega^\perp$, make the surface measure dy. They immediately give

$$Rf(-s,-\omega) = Rf(s,\omega), \quad R(\partial_j f) = \omega_j \partial_s Rf, \quad R(\Delta f) = \partial_s^2 Rf.$$

For the middle identity decompose ∂_j into a normal and tangential derivative. Tangential integrals vanish by compact support. Induction gives $R(D^\gamma f) = \omega^\gamma \partial_s^{|\gamma|} Rf$. If $\mathrm{supp} f \subset B_R$, then $Rf = 0$ for $|s| \ge R$.



$$Rf(s,\omega) = \int_{x\cdot\omega=s} f\, dS$$

fix the unit normal ω;
vary s to scan parallel lines

Diagram T4.3. Radon data integrate along level hyperplanes. In dimension two the integration sets are lines perpendicular to ω. The scalar s is signed distance in the normal direction, not distance traveled along a line. The blue line is one member of the family; higher dimensions replace it by an $(n-1)$-plane.

Theorem 4.7 (Slice theorem and inversion; source Theorems 4–5)
With the unnormalized one-dimensional transform $B(r,\omega) = \int_{\mathbb{R}} e^{-irs} Rf(s,\omega)\, ds$,

$$B(r,\omega) = (2\pi)^{n/2}\hat{f}(r\omega).$$

For every dimension,

$$f(x) = \frac{1}{2(2\pi)^n}\int_{S^{n-1}}\int_{\mathbb{R}} B(r,\omega)|r|^{n-1}e^{irx\cdot\omega}\, dr\, dS_\omega.$$

If $n = 2k+1$ *is odd, this becomes the local formula*

$$f(x) = C_n \int_{S^{n-1}} \partial_s^{2k} Rf(x\cdot\omega,\omega)\, dS_\omega, \qquad C_n = \frac{(-1)^k}{2(2\pi)^{2k}}. \quad \text{(RI4)}$$

Expanded proof.

Fubini in the orthonormal coordinates gives

$$B(r,\omega) = \int f(x)e^{-ir\omega\cdot x}\, dx.$$

In Fourier inversion, polar coordinates with signed r double-count each nonzero frequency, giving the factor $1/2$ and $|r|^{n-1}$. All integrals converge since $\hat{f}$ is rapidly decreasing. For $n-1 = 2k$, one-dimensional inversion gives

$$\partial_s^{2k} Rf(s,\omega) = \frac{(-1)^k}{2\pi}\int_{\mathbb{R}} r^{2k}B(r,\omega)e^{irs}\, dr.$$

Substituting this in the general inversion formula proves (RI4). □

For odd n, define $\rho_f = C_n \partial_s^{n-1} Rf$. Then the plane-wave integral

$$u(x,t) = \int_{S^{n-1}} \rho_g(x\cdot\omega - t, \omega)\, dS_\omega$$

solves $u_{tt} - \Delta u = 0, u(0) = g, u_t(0) = 0$. The PDE holds because $|\omega| = 1$; the first initial condition is (RI4). The integrand for $u_t(0)$ is odd under $\omega \mapsto -\omega$, since an odd s derivative reverses the symmetry of Rg.

For the constant-coefficient system $\mathbf{u}_t + \sum B_j \mathbf{u}_{x_j} = 0$, suppose $B(\omega) = \sum \omega_j B_j$ is diagonalizable with real eigenvalues $\lambda_l(\omega)$. In Radon coordinates the diagonal components satisfy one-dimensional transport and equal

$$v_l(s,\omega,t) = h_l(s - \lambda_l(\omega)t, \omega).$$

After applying (RI4), use $\rho_l(s,\omega) = C_n \partial_s^{n-1} h_l(s,\omega)$ exactly once:

$$\mathbf{u}(x,t) = \int_{S^{n-1}} \sum_l a_l(\omega)\, \rho_l(x\cdot\omega - \lambda_l(\omega)t, \omega)\, dS_\omega.$$

Here a_l are the diagonalizing columns; the shift is not inserted again in the definition of ρ_l. For compact data in B_R, if $\min_{\omega,l} |\lambda_l(\omega)| = \sigma > 0$, then $\mathbf{u} = 0$ for $|x| < \sigma t - R$. Indeed each shifted argument has magnitude at least $\sigma t - |x| > R$. This is a conditional odd-dimensional Huygens statement; the diagonalization and nonzero-speed hypotheses cannot be removed.

奇偶维差别藏在绝对值里

带符号的极径在一般维数必须用 $|r|^{n-1}$；只有奇维时才等于多项式 r^{2k}，从而变成有限阶局部导数。偶维的反演通常带非局部算子。

4.3.4 Laplace transform, resolvents and the heat–wave link

If v and v_t have suitable exponential bounds, integration by parts gives

$$\mathscr{L}(v_t)(s) = s\mathscr{L}v(s) - v(0), \qquad \mathscr{L}v(s) = \int_0^\infty e^{-st} v(t)\, dt.$$

Thus for $v_t - \Delta v = 0$, $v(0) = f$,

$$(s - \Delta)\mathscr{L}v = f.$$

Use the same spatial boundary condition before and after transformation; for example a zero Dirichlet condition stays zero. In the whole space this recovers $(s-\Delta)^{-1} f = \int_0^\infty e^{-st}(G_t * f)\, dt$. For L^2 data the Bochner integral is justified by $\|G_t * f\|_2 \le \|f\|_2$.

平面支集条件怎样变成内部空区

若 $|x| < \sigma t - R$，所有方向的平面参数都移出初始支集区间，整个反演积分为零。零特征速度会留下不移动的分量，因此必须单独排除。

Theorem 4.8 (Gaussian averaging of the wave group; source Example 9)
Let g be smooth compactly supported, and let $U(x,s)$ be the wave solution with displacement g and zero initial velocity, extended evenly in s. Then

$$V(x,t) = \frac{1}{\sqrt{4\pi t}} \int_{\mathbb{R}} e^{-s^2/(4t)} U(x,s)\, ds$$

is the heat solution with datum g.

Expanded proof.

For smooth data the derivatives of U grow at most polynomially, as can also be checked from its Fourier representation. Gaussian weights justify differentiation and kill the boundary terms. Writing $q_t(s) = (4\pi t)^{-1/2} e^{-s^2/(4t)}$, observe $\partial_t q_t = \partial_{ss} q_t$. Hence

$$V_t = \int q_{ss} U = \int q U_{ss} = \int q \Delta U = \Delta V.$$

The Gaussian approximate identity in s gives $V(0) = g$. Alternatively the frequency multiplier is $\int q_t(s) \cos(s|\xi|)\, ds = e^{-t|\xi|^2}$, which proves the same statement in L^2. □

For $n = 2k+1 \geq 3$, let $M_g(x,r)$ be the spherical average of g. Equating the two Gaussian representations gives, with $\lambda = 1/(4t)$,

$$\int_0^\infty e^{-\lambda s^2} U(x,s)\, ds = \frac{|S^{n-1}|}{2\pi^k} \lambda^k \int_0^\infty e^{-\lambda r^2} r^{2k} M_g(x,r)\, dr.$$

Use $\lambda e^{-\lambda r^2} = -(2r)^{-1} \partial_r e^{-\lambda r^2}$ and integrate by parts k times. The result is

$$\lambda^k \int_0^\infty e^{-\lambda r^2} r^{2k} M_g\, dr = 2^{-k} \int_0^\infty e^{-\lambda r^2} r (r^{-1}\partial_r)^k (r^{2k-1} M_g)\, dr.$$

There are no boundary terms at infinity by compact support in r. At zero, M_g is smooth and even; before the last differentiation the relevant odd power has not fallen below one, so the boundary terms are zero. After $s^2 = \tau$, uniqueness of the Laplace transform implies

$$U(x,t) = \frac{1}{(n-2)!!} \partial_t (t^{-1}\partial_t)^{(n-3)/2} (t^{n-2} M_g(x,t)).$$

Here $|S^{n-1}|/(2^{k+1}\pi^k) = 1/(n-2)!!$ follows from the Gaussian sphere area formula. The one-dimensional case is $U = (g(x+t) + g(x-t))/2$. One can justify Laplace uniqueness without a new PDE theorem: after multiplying by $e^{-a\tau}$, substitute $z = e^{-\tau}$; vanishing at all $s = a + j$ gives zero moments on $[0,1]$. Polynomial density in continuous functions makes the finite signed measure zero. This applies to the continuous, exponentially integrable functions above.

初值正是 Laplace 分部积分的边界项

时间导数并非只替换成乘以 s。零时刻留下 $-v(0)$，它成为椭圆 resolvent 方程的右端；漏掉这一项就把非零初值问题变成了齐次问题。

4.4 Converting Nonlinear into Linear PDE

4.4.1 Cole–Hopf: choose the change of unknown by cancellation

Consider $u_t - a\Delta u + b|Du|^2 = 0$, $a > 0$. For $w = \phi(u)$, the chain rule yields

$$w_t - a\Delta w = -(b\phi'(u) + a\phi''(u))|Du|^2.$$

To cancel the nonlinear term solve $a\phi'' + b\phi' = 0$. For $b \neq 0$ choose $\phi(z) = e^{-bz/a}$.

Theorem 4.9 (Cole–Hopf; source §4.4.1)
If g is bounded and continuous, define

$$w(x,t) = \int_{\mathbb{R}^n} G_{at}(x-y)e^{-bg(y)/a}\,dy, \qquad u = -\frac{a}{b}\log w. \tag{CH4}$$

Then u is smooth for $t > 0$, solves the nonlinear PDE, and has initial value g at continuity points. It is the unique bounded classical solution with these data in the corresponding heat uniqueness class.

Expanded proof.

Boundedness of g gives positive constants $0 < c \le w \le C$, since the heat kernel is positive and has mass one. Heat smoothing, differentiation under Gaussian integrals and $w_t = a\Delta w$ hold for $t > 0$. Thus the logarithm is well defined and differentiable; substitution or the preceding chain rule proves the nonlinear equation. The approximate-identity limit gives $w \to e^{-bg/a}$ and hence $u \to g$. Conversely any bounded classical solution transforms into a bounded heat solution. Heat uniqueness and injectivity of the exponential prove the assertion. If $b = 0$, the original equation is already the heat equation; division by b in (CH4) is not allowed. □

For viscous Burgers, $u_t + uu_x - au_{xx} = 0$, let $P' = g$. A potential W with $W_x = u$ satisfies $W_t + \frac{1}{2}W_x^2 - aW_{xx} = c(t)$. Subtracting $\int c$ removes this purely time-dependent term. With $Z = e^{-W/(2a)}$, $Z_t = aZ_{xx}$, so

$$W = -2a\log\int_{\mathbb{R}} G_{at}(x-y)e^{-P(y)/(2a)}\,dy,$$

$$u(x,t) = \frac{\int_{\mathbb{R}}(x-y)t^{-1}e^{-\{(x-y)^2/(2t)+P(y)\}/(2a)}\,dy}{\int_{\mathbb{R}}e^{-\{(x-y)^2/(2t)+P(y)\}/(2a)}\,dy}. \tag{VB4}$$

For bounded g, choose $P(x) = \int_0^x g$; it is Lipschitz and need not be bounded. The Gaussian dominates its at-most-linear growth, so the integrals and their derivatives on compact positive-time sets converge. Integrating the y derivative of the exponential shows also u is its weighted

正性同时保证变换可逆

指数变换总是正的，而热核保持严格正性。反变换要取对数，所以不能把任意带符号的热解代入；有界初值还给出远离零的统一下界。

average of $g(y)$, a.e.:

$$\int_{\mathbb{R}} \big((x-y)/t - g(y)\big) e^{-K(y)/(2a)}\, dy = 0.$$

The endpoint terms vanish by quadratic Gaussian decay. In particular $|u| \le \|g\|_\infty$.

4.4.2 Potential flow and Bernoulli's law

For incompressible Euler, $\mathbf{u}_t + (\mathbf{u}\cdot D)\mathbf{u} = -Dp + Dh$ and $\operatorname{div}\mathbf{u} = 0$, impose the additional ansatz $\mathbf{u} = Dv$. Then $\Delta v = 0$, and componentwise symmetry of $D^2 v$ gives

$$[(\mathbf{u}\cdot D)\mathbf{u}]_i = \sum_j v_{x_j} v_{x_i x_j} = \partial_i \left(\tfrac{1}{2}|Dv|^2\right).$$

Therefore

$$D(v_t + \tfrac{1}{2}|Dv|^2 + p - h) = 0, \qquad p = h - v_t - \tfrac{1}{2}|Dv|^2 + c(t)$$

on each connected spatial component. To match initial data one needs $\mathbf{g} = Dv_0$ as well as $\operatorname{div}\mathbf{g} = 0$. Locally curl-free data admit a potential; globally a topological condition or a prescribed global potential is required.

4.4.3 Hodograph: invert dependent and independent variables

Write the two velocity components as $p = u^1$, $q = u^2$. For steady irrotational compressible flow the equations are

$$(c^2 - p^2)p_{x_1} - pq(p_{x_2} + q_{x_1}) + (c^2 - q^2)q_{x_2} = 0, \qquad p_{x_2} = q_{x_1},$$

where $c = c(p,q) > 0$ is given. If $J = \det D_x(p,q) \neq 0$, invert to $x = X(p,q)$. The inverse matrix identity is

$$\begin{pmatrix} p_{x_1} & p_{x_2} \\ q_{x_1} & q_{x_2} \end{pmatrix} = J \begin{pmatrix} X^2_q & -X^1_q \\ -X^2_p & X^1_p \end{pmatrix}.$$

Substitution and division by J give the linear system

$$(c^2 - p^2)X^2_q + pq(X^1_q + X^2_p) + (c^2 - q^2)X^1_p = 0, \qquad X^1_q = X^2_p.$$

Locally $X = DV$ for a scalar potential $V(p,q)$, and

$$(c^2 - q^2)V_{pp} + 2pqV_{pq} + (c^2 - p^2)V_{qq} = 0.$$

The coefficient determinant is $c^2(c^2 - p^2 - q^2)$: the equation is elliptic in the subsonic region and hyperbolic in the supersonic region, degenerate at sonic speed.

原函数的常数与时间规范

给 P 加常数只会把热解 Z 乘一个正数，不改变 $u = -2aZ_x/Z$。从 Burgers 积分得到的时间函数也能吸收进势函数，不能无说明地直接丢掉。

无散初值还不足以成为势流

$\operatorname{div}\mathbf{g} = 0$ 是不可压条件，不能推出 $\mathbf{g} = Dv_0$。还需无旋及相应的全局条件；这里线性化的是势流子类，并未解出一般 Euler 方程。

4.4.4 A local Legendre transform for the minimal surface equation

In two dimensions the minimal surface equation is

$$(1+u_{x_2}^2)u_{x_1x_1} - 2u_{x_1}u_{x_2}u_{x_1x_2} + (1+u_{x_1}^2)u_{x_2x_2} = 0.$$

Assume $\det D^2u = J \neq 0$ locally. Use $p = Du(x)$ as independent variable, let $x = X(p)$, and set $v(p) = X(p)\cdot p - u(X(p))$.

Expanded proof (the Hessian inverse and the linear equation).

The chain rule gives

$$\partial_{p_i} v = X_i + \sum_j p_j \partial_{p_i} X_j - \sum_j u_{x_j}\partial_{p_i}X_j = X_i.$$

Hence $D^2v = D_pX = (D^2u)^{-1}$. For a symmetric 2×2 matrix the inverse formula implies

$$u_{x_1x_1} = Jv_{p_2p_2}, \quad u_{x_1x_2} = -Jv_{p_1p_2}, \quad u_{x_2x_2} = Jv_{p_1p_1}.$$

Substitute and divide by J to obtain

$$(1+p_1^2)v_{p_1p_1} + 2p_1p_2v_{p_1p_2} + (1+p_2^2)v_{p_2p_2} = 0.$$

Its coefficient matrix $I + p\otimes p$ is positive definite. Conversely, a solution with invertible Hessian yields a local inverse $x = Dv(p)$ and reconstructs $u = x\cdot p - v$ satisfying the original PDE. Boundary data need a separate transformation; a fixed boundary in x generally becomes an unknown surface in p. □

变线性没有消除退化

系数现在依赖独立变量，因而方程线性，但声速边界仍使主部退化。还要把 $X(p,q)$ 反解回去；Jacobian 变零处不能继续套用变换。

4.5 Asymptotics

The source presents several asymptotic constructions formally. We distinguish exact algebraic identities, proved local integral asymptotics, and conditional PDE expansions. No convergence theorem is inferred merely by matching powers.

局部 Legendre 变换不要求凸性

这里凭 $\det D^2u \neq 0$ 局部反演，Hessian 可以不定。第三章的凸共轭是全局上确界构造；两者在严格凸、可逆情形相接，但假设与用途不同。

4.5.1 A small-diffusion plume: geometry and transverse scale

Consider $\operatorname{div}(\mathbf{b}\,\mu) = \delta_0$ in two dimensions. Let $\dot X = \mathbf{b}(X)$, $X(0) = 0$, and suppose $|\mathbf{b}(X)| > 0$. If the forward trajectory leaves compact sets, the locally finite measure

$$\int \phi\, d\mu = \int_0^\infty \phi(X(t))\, dt$$

solves this equation, since

$$-\int \mathbf{b}\cdot D\phi\, d\mu = -\int_0^\infty \frac{d}{dt}\phi(X(t))\, dt = \phi(0).$$

This is an outgoing solution; an extra background homogeneous solution is not excluded by this calculation. With speed $\sigma = |\mathbf{b}(X)|$, tangent τ and unit normal ν, use $x = X(t) + y\nu(t)$ in a nonoverlapping tubular neighborhood. Frenet identities give $\nu' = -\sigma\kappa\tau$ and

$$x_t = \sigma(1-\kappa y)\tau, \quad x_y = \nu, \quad dx = H\, dt\, dy, \qquad H = \sigma(1-\kappa y) > 0.$$

Thus the same measure has density $\delta(y)/\sigma(t)$ with respect to spatial area.

Now perturb to $-\varepsilon\Delta u^\varepsilon + \operatorname{div}(\mathbf{b}u^\varepsilon) = \delta_0$. Away from the injection point, set $z = y/\sqrt{\varepsilon}$, $u^\varepsilon = \varepsilon^{-1/2}v^\varepsilon(t,z)$. The coordinate metric gives the exact identity

$$\Delta w = \frac{1}{H}\partial_t(H^{-1}w_t) + \frac{1}{H}\partial_y(Hw_y).$$

Also write $\mathbf{b} = (\sigma + \alpha y + O(y^2))\tau + (\beta y + O(y^2))\nu$. Then on compact $t > 0$, bounded z, provided the needed derivatives of v^ε are uniformly bounded,

$$\mathbf{b}\cdot Dv^\varepsilon = v_t^\varepsilon + \beta z v_z^\varepsilon + O(\sqrt{\varepsilon}), \qquad \varepsilon\Delta v^\varepsilon = v_{zz}^\varepsilon + O(\sqrt{\varepsilon}),$$

$$\operatorname{div}\mathbf{b} = \sigma'/\sigma + \beta + O(\sqrt{\varepsilon}).$$

For example the transverse first-derivative term in the Laplacian is $\varepsilon(H_y/H)\partial_y v^\varepsilon = O(\sqrt{\varepsilon})$; the tangential second-derivative terms are $O(\varepsilon)$. Thus the conditional leading equation is

$$v_t - v_{zz} + (\beta z v)_z + (\sigma'/\sigma)v = 0, \qquad v(0) = \delta_0/\sigma(0). \tag{PL4}$$

Its exact Gaussian solution, derived again in Problem 13, is

$$v(t,z) = \frac{e^{-z^2/\Gamma(t)}}{\sigma(t)\sqrt{\pi\Gamma(t)}}, \qquad \Gamma' = 4 + 2\beta\Gamma, \quad \Gamma(0) = 0.$$

Indeed $A = \sigma v$ satisfies $A_t = A_{zz} - (\beta z A)_z$; substituting the Gaussian gives the displayed ODE and mass $\int A = 1$. The transverse width is $\sqrt{\varepsilon\Gamma(t)}$. This does not by itself prove approximation of the original elliptic Green function, nor control the injection region or large times.

线上的密度必须带速度因子

注入量按行进时间计量，而面积元在曲线处含 $\sigma(t)$。因此横向 Dirac 前要乘 $1/\sigma$。若轨道封闭或反复回返，时间积分可能不是局部有限测度。

4.5.2 Laplace concentration and vanishing viscosity

Lemma 4.10 (Source asymptotics lemma, with tail control)
Let k, l be continuous on $\mathbb{R}$, $|l(y)| \le C(1+|y|)$, and $k(y) \ge cy^2 - C$, $c > 0$. If k has a unique minimum at y_0, then

$$\frac{\int l(y) e^{-k(y)/\varepsilon}\, dy}{\int e^{-k(y)/\varepsilon}\, dy} \longrightarrow l(y_0). \tag{LC4}$$

Expanded proof.

Subtract the minimum k_0 and put $h = k - k_0 \ge 0$. Fix a small neighborhood N of y_0 on which $|l - l(y_0)| < \eta$. On a large compact interval outside N, $h \ge d > 0$ by uniqueness. Increase the interval so that outside it $h(y) \ge d/2 + (c/2)y^2$, decreasing d if necessary. By continuity there is an interval of length $2r > 0$ near y_0 where $h < d/4$. The denominator is at least $2re^{-d/(4\varepsilon)}$. The numerator of the absolute error over the compact complement is at most $C_1 e^{-d/\varepsilon}$. The tail numerator is bounded by

$$C_2 e^{-d/(2\varepsilon)} \int (1+|y|) e^{-cy^2/(2\varepsilon)}\, dy,$$

whose ratio to the denominator tends to zero. Inside N the normalized error is at most η. Let $\varepsilon \downarrow 0$ and then $\eta \downarrow 0$. This proves concentration together with the required control of the unbounded observable l. □

For bounded initial g, let $P' = g$ and $K(y) = (x-y)^2/(2t) + P(y)$. The quadratic term dominates the Lipschitz primitive. Formula (VB4) and (LC4), with parameter 2ε, imply

$$u^\varepsilon(x,t) \longrightarrow \frac{x - y(x,t)}{t}$$

whenever the minimizing point is unique. By Chapter 3, at each $t > 0$ only countably many x have multiple minimizers, and the right side is the Lax–Oleinik entropy solution. Since $|u^\varepsilon| \le \|g\|_\infty$, this convergence is also L^1_{loc} in x for each fixed t. Joint a.e. convergence and the same bound give local space-time L^1 convergence away from zero. The multiple-minimizer set is measurable, for instance by expressing separation of two minimizers using rational compact neighborhoods; Fubini therefore applies to the countable time slices.

形式余项依赖尚未证明的统一界

把 y 换成 $\sqrt{\varepsilon} z$ 后，系数误差很小，但它仍乘着解的导数。只有这些导数在所考察区域有统一界，才能写成统一的 $O(\sqrt{\varepsilon})$。

逐点密度趋零不足以推出集中

归一化密度在每个非极小点趋零，还不能控制积分或无穷远的质量。证明需要分母下界、远离极小点的能量间隙，以及带权尾积分。

If k is smooth with finitely many nondegenerate global minima y_j, local Taylor expansion $y = y_j + \sqrt{\varepsilon} z$ and Gaussian domination give

$$\int l e^{-k/\varepsilon} = e^{-k_0/\varepsilon} \sqrt{2\pi\varepsilon} \left(\sum_j \frac{l(y_j)}{\sqrt{k''(y_j)}} + o(1) \right).$$

To justify domination, restrict to a small neighborhood where $k - k_0 \geq c_j |y - y_j|^2$, and use the energy gap outside. Thus at a tie the limit is a weighted average, not necessarily the velocity at one selected minimizing point. Values on shock curves do not affect the weak solution.

与第三章的最小化公式接上了

黏性解是最优速度的指数加权平均。黏性趋零时，权重集中到作用量最低点；最低点的分支竞争正对应守恒律的激波。

4.5.3 Geometric optics and stationary phase

For $u^\varepsilon = a e^{ip/\varepsilon}$ the exact wave-operator identity is

$$\begin{aligned} & e^{-ip/\varepsilon} (u^\varepsilon_{tt} - \Delta u^\varepsilon) \\ & \quad = \varepsilon^{-2} a (|Dp|^2 - p_t^2) \\ & \qquad + \frac{i}{\varepsilon} \{ 2 p_t a_t - 2 Dp \cdot Da + a(p_{tt} - \Delta p) \} + (a_{tt} - \Delta a). \end{aligned}$$

Matching the first two orders gives the eikonal and amplitude equations

$$p_t^2 = |Dp|^2, \qquad 2 p_t a_t - 2 Dp \cdot Da + (p_{tt} - \Delta p) a = 0.$$

On a smooth nonzero-frequency branch choose $p_t = \pm |Dp|$. The sign cannot be chosen independently at every point. For $u_{tt} - \sum a_{ij}(x) u_{x_i x_j} = 0$ the first equation becomes $p_t^2 = \sum a_{ij} p_{x_i} p_{x_j}$, with the corresponding amplitude expression obtained by the same differentiation. These identities define a candidate expansion. A theorem about exact solutions needs control of the residual and the evolution, especially near caustics.

> **Lemma 4.11** (Nonstationary phase; strengthening source Lemma 1)
> *If $a \in C_c^\infty$ and $|D\phi| > 0$ on its support, then for every integer $N \geq 1$, $\int e^{i\phi/\varepsilon} a = O(\varepsilon^N)$.*

Expanded proof.

Near the support let $V = D\phi / |D\phi|^2$. Then $V \cdot D e^{i\phi/\varepsilon} = (i/\varepsilon) e^{i\phi/\varepsilon}$. Integration by parts gives

$$\int e^{i\phi/\varepsilon} a = -\frac{\varepsilon}{i} \int e^{i\phi/\varepsilon} \operatorname{div}(aV).$$

Iterate N times; the final smooth compactly supported amplitude has finite L^1 norm independent of ε. This also proves the source's linear-phase case. □

相位误差除以小参数会放大

即使 $p^\varepsilon - p = o(1)$，也不能直接推出两个指数相近；指数里的差是 $(p^\varepsilon - p)/\varepsilon$。振荡展开需要比普通逐点收敛更精确的误差尺度。

Lemma 4.12 (Quadratic phase; source Lemma 2)
For real symmetric invertible H and $a \in C_c^\infty(\mathbb{R}^d)$,

$$\int e^{iz\cdot Hz/(2\varepsilon)} a(z)\,dz = (2\pi\varepsilon)^{d/2} \frac{e^{i\pi\,\operatorname{sgn} H/4}}{|\det H|^{1/2}} [a(0) + O(\varepsilon)]. \quad \text{(QP4)}$$

Expanded proof (regularization, branch and error).

For $\operatorname{Re} q > 0$,

$$\int_{\mathbb{R}} e^{-qs^2+ibs}\,ds = \sqrt{\pi}\, q^{-1/2} e^{-b^2/(4q)}.$$

Choose the square root with positive real part. For real positive q this is the Gaussian identity; analytic continuation in the right half-plane proves it for complex q, since compact subsets admit an integrable Gaussian majorant. Equivalently the contour argument of Problem 15 proves it directly.

Diagonalize H orthogonally, multiply the phase by $e^{-\eta|z|^2}$, and apply this identity in each coordinate. As $\eta \downarrow 0$, an eigenvalue λ contributes the factor $|\lambda|^{-1/2} e^{i\pi\,\operatorname{sgn}\lambda/4}$ after the common $(2\pi\varepsilon)^{1/2}$ factor. Insert Fourier inversion for a before this calculation. Its transform is rapidly decreasing. The regularized Gaussian integral is uniformly bounded in frequency, so dominated convergence yields the exact oscillatory identity

$$\int e^{iz\cdot Hz/(2\varepsilon)} a(z)\,dz = C_{\varepsilon,H}(2\pi)^{-d/2} \cdot \int \hat{a}(\xi) e^{-i\varepsilon\,\xi\cdot H^{-1}\xi/2}\,d\xi,$$

where $C_{\varepsilon,H}$ is the prefactor in (QP4). Since $|e^{ir} - 1| \le |r|$, the bracket differs from $a(0)$ by at most

$$C\varepsilon \int |\xi|^2 |\hat{a}(\xi)|\,d\xi < \infty.$$

This proves both the constant and the error in (QP4). □

Lemma 4.13 (Morse coordinates; source Lemma 3(ii))
If $D\phi(0) = 0$ and $H = D^2\phi(0)$ is invertible, there is a local smooth diffeomorphism Φ with $\Phi(0) = 0$, $D\Phi(0) = I$, and $\phi(\Phi(z)) = \phi(0) + z\cdot Hz/2$.

Expanded proof (an editorial matrix proof).

Taylor's formula with integral remainder gives $\phi(x) = \phi(0) + x^T A(x) x/2$, where $A(x) = 2\int_0^1 (1-s) D^2\phi(sx)\,ds$ and $A(0) = H$. On symmetric matrices solve

$$H + 2S + SH^{-1}S = A(x).$$

The derivative in S at zero is $2I$, so the implicit function theorem gives a smooth symmetric $S(x)$ with $S(0) = 0$. Set $B = I + H^{-1}S$; then $B^T H B = A$. The map $z = B(x)x$ has derivative I at zero. Its local inverse Φ has the required properties. This works for indefinite H as well as positive definite H. □

For completeness the source's noncritical coordinate lemma is also elementary. Choose rows $r_1, \dots, r_{d-1}$ complementary to $D\phi(0)$, impose $r_j \cdot (y - x) = 0$ and $\phi(y) - \phi(0) = D\phi(0) \cdot x$. The y Jacobian is invertible; the implicit function theorem gives $y = \Phi(x)$ and differentiation gives $D\Phi(0) = I$.

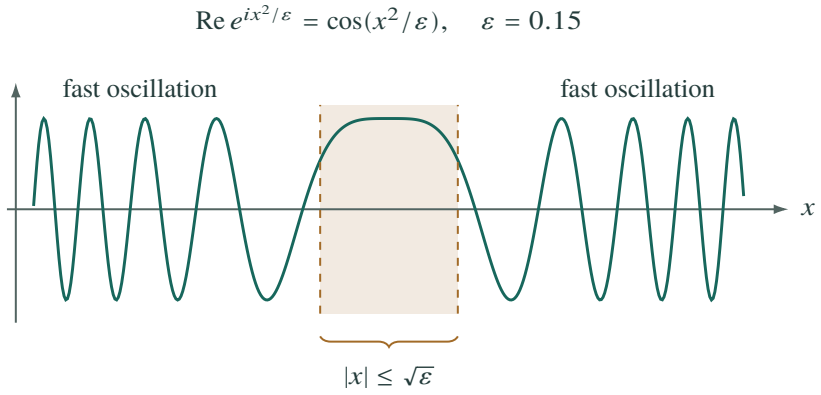


Diagram T4.4. Near a nondegenerate stationary point the scale is square-root. For phase x^2, oscillation is slow near its stationary point at zero and rapid farther away. The highlighted interval has width $2\sqrt{\varepsilon}$. Rescaling $x = \sqrt{\varepsilon}\, z$ explains the leading $\varepsilon^{1/2}$ factor in one dimension; cancellation, not pointwise smallness of the integrand, suppresses the outer region.

Theorem 4.14 (Nondegenerate stationary phase)
Suppose ϕ has just finitely many critical points z_j on supp a*, and all $H_j = D^2\phi(z_j)$ are invertible. Then*

$$\int e^{i\phi(z)/\varepsilon} a(z)\, dz = (2\pi\varepsilon)^{d/2} \sum_j \frac{e^{i\phi(z_j)/\varepsilon + i\pi\, \operatorname{sgn} H_j/4}}{|\det H_j|^{1/2}} [a(z_j) + O(\varepsilon)]. \tag{SP4}$$

Expanded proof.

Choose a partition of unity separating small neighborhoods of the critical points from the noncritical compact remainder. The latter is $O(\varepsilon^N)$ for arbitrary N by the first lemma. At each critical point change to Morse coordinates. The transformed amplitude is $a(\Phi(z))|\det D\Phi(z)|$ times its cutoff and equals $a(z_j)$ at the origin. Apply (QP4) to each localized integral and add them. Taking $N > d/2 + 1$ absorbs the noncritical remain-

der. □

Recovering the rays of Chapter 3. For high-frequency displacement $g^\varepsilon(z) = a(z)e^{ip_0(z)/\varepsilon}$ with $|Dp_0| > 0$ on $\operatorname{supp} a$ and zero initial velocity, Fourier representation splits the wave into two equal half-waves. After frequency rescaling their phases are

$$\phi_\pm(x, y, z, t) = (x - z) \cdot y \pm t|y| + p_0(z).$$

The critical equations are

$$y = Dp_0(z), \qquad x = z \mp t\,\frac{Dp_0(z)}{|Dp_0(z)|}. \tag{RAY4}$$

These are exactly the projected HJ rays for $p_t \mp |Dp| = 0$. The joint Hessian at a critical point is

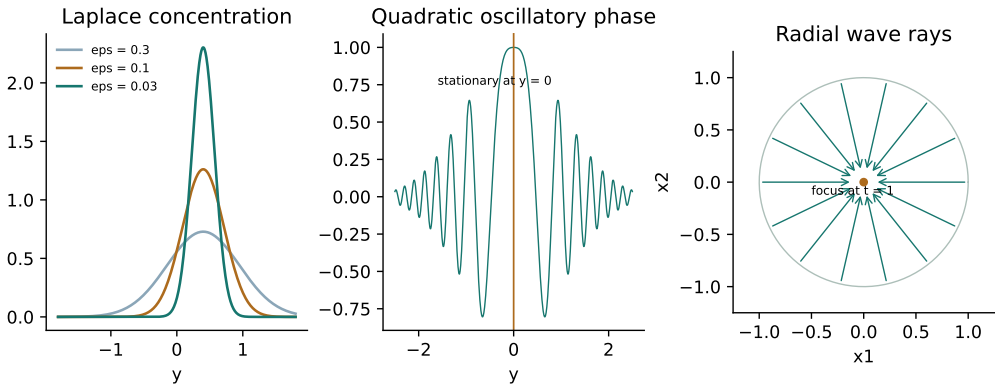
$$\begin{pmatrix} \pm\dfrac{t}{|y|}(I - \hat{y} \otimes \hat{y}) & -I \\ -I & D^2 p_0(z) \end{pmatrix}, \qquad \hat{y} = y/|y|.$$

Its determinant equals $(-1)^n$ times the Jacobian determinant of the ray map in (RAY4), by block elimination. For small time that map is locally invertible. The critical value is $p_0(z)$, since $(x-z)\cdot y \pm t|y| = 0$. Applying (SP4) in $2n$ variables cancels the half-wave prefactor $(2\pi\varepsilon)^{-n}$ and gives the amplitude $a(z)/|\det D_z x|^{1/2}$ and the Hessian signature phase. The two half-waves are then added with weight $1/2$.

The amplitude in y, z is initially not compact in y. To justify this application, first cut frequencies off to a compact neighborhood of $Dp_0(\operatorname{supp} a)$ disjoint from zero. Outside that neighborhood $D_z\phi = Dp_0 - y$ is nonzero. Repeated integration by parts in z makes the tail integrable in y, with arbitrarily high powers of ε; for large y each integration also supplies inverse powers of $|y|$. This reduces to the compactly supported theorem. The formula is local before caustics or summed over the finitely many nondegenerate preimages; it asserts no uniform approximation at a caustic.

行列式为零时不能继续用驻相公式

$|\det H|^{-1/2}$ 描述非退化临界点的集中强度。它发散时，表示这一阶近似失效；需研究更高阶相位，不能把发散系数直接当作精确解的无穷大。



Editorial figure. Laplace concentration is compared with oscillatory cancellation; the ray diagram displays a focusing map whose Jacobian eventually vanishes.

4.5.4 Periodic homogenization: expansion, cell problem, and a proof

We use the symmetric uniformly elliptic case, sufficient also for Problem 18: $A(y) = A(y)^T$ is smooth, Q-periodic, $Q = (0,1)^n$, and $\theta I \le A \le \Theta I$. Consider $-\operatorname{div}(A(x/\varepsilon)Du^\varepsilon) = f$ on a bounded Lipschitz domain with zero trace and $f \in L^2$. The source uses a formal expansion; we first derive it, then give a weak-convergence proof under these stated hypotheses.

Two-scale bookkeeping. Write $u^\varepsilon = u_0(x,y) + \varepsilon u_1(x,y) + \cdots$, $y = x/\varepsilon$, with all u_j periodic in y. Every spatial derivative becomes $D_x + \varepsilon^{-1} D_y$. Thus the three operator pieces are

$$L_1 = -D_y \cdot AD_y, \quad L_2 = -D_y \cdot AD_x - D_x \cdot AD_y, \quad L_3 = -D_x \cdot AD_x.$$

The coefficients of $\varepsilon^{-2}, \varepsilon^{-1}, 1$ give

$$L_1 u_0 = 0, \quad L_1 u_1 + L_2 u_0 = 0, \quad L_1 u_2 + L_2 u_1 + L_3 u_0 = f.$$

Testing the first identity by u_0 and integrating over Q gives $D_y u_0 = 0$; opposite periodic faces cancel. For zero-mean correctors χ_i, define

$$\operatorname{div}_y\{A(y)(e_i - D_y\chi_i)\} = 0, \qquad \int_Q \chi_i = 0. \tag{CELL4}$$

Then $u_1 = -\sum_i \chi_i(y) u_{x_i}$ solves the next equation. Averaging the order-one equation kills $L_1 u_2$ and yields

$$-\operatorname{div}(\overline{A}Du) = f, \qquad \overline{A}e_i = \int_Q A(y)(e_i - D\chi_i)\,dy. \tag{HOM4}$$

The minus sign in u_1 matches the source's corrector convention. Many other texts use $\varphi_i = -\chi_i$ and a plus sign instead.

Lemma 4.15 (Cell solvability and effective ellipticity)
(CELL4) has a unique zero-mean periodic weak solution. The matrix $\overline{A}$ is symmetric and satisfies $\theta I \le \overline{A} \le \Theta I$.

Expanded proof.

On the zero-mean periodic H^1 space minimize $\int_Q (e_i - Dv)\cdot A(e_i - Dv)$. Periodic Poincaré and ellipticity bound every minimizing sequence in H^1; weak compactness and weak lower semicontinuity give a minimizer. The first variation is $\int_Q A(D\chi_i - e_i)\cdot D\psi = 0$. Strict convexity in the gradient makes the difference of two minimizers constant, hence zero after normalization. These are standard Hilbert-space compactness and periodic Poincaré facts; the cell equation is not solved merely by asserting zero mean.

For $\xi \in \mathbb{R}^n$, let $\chi_\xi = \sum_i \xi_i \chi_i$. The cell identity allows any gradient $D\chi_\eta$ to be subtracted from the other factor, so

$$\eta \cdot \overline{A}\xi = \int_Q (\eta - D\chi_\eta)\cdot A(\xi - D\chi_\xi).$$

This proves symmetry. The minimum over all periodic v equals $\xi\cdot\overline{A}\xi$ by orthogonality. Its lower bound is $\theta\int_Q |\xi - Dv|^2 \ge \theta|\xi|^2$, since $\int_Q Dv = 0$ and $|Q| = 1$. The competitor $v = 0$ gives the upper bound. □

Theorem 4.16 (An editorial rigorous homogenization statement)
Under the symmetric hypotheses above, $u^\varepsilon \rightharpoonup u$ in H_0^1 and $u^\varepsilon \to u$ in L^2, where u is the unique weak solution of (HOM4).

Expanded proof (oscillating tests replace an unproved expansion).

Testing the original PDE by u^ε gives a uniform H_0^1 bound by ellipticity and Poincaré. Take a weakly convergent subsequence and use Rellich for strong L^2 convergence. The fluxes $\eta^\varepsilon = A(x/\varepsilon)Du^\varepsilon$ are bounded in L^2; take a weak limit η. The PDE gives $-\operatorname{div}\eta = f$.

For each i put

$$w_i^\varepsilon = x_i - \varepsilon\chi_i(x/\varepsilon), \qquad q_i^\varepsilon = A(x/\varepsilon)Dw_i^\varepsilon.$$

Periodic averaging implies $w_i^\varepsilon \to x_i$ in L^2_{loc} and $q_i^\varepsilon \rightharpoonup \overline{A}e_i$ in L^2_{loc}; also $\operatorname{div} q_i^\varepsilon = 0$. The averaging fact follows first by summing integrals on full ε-cells for continuous periodic functions and compact tests, then by $L^2(Q)$ approximation and uniform local cell bounds.

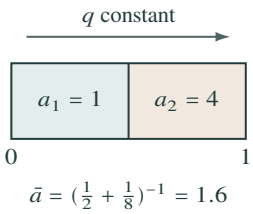
For $\phi \in C_c^\infty(U)$ symmetry gives $\eta^\varepsilon \cdot Dw_i^\varepsilon = Du^\varepsilon \cdot q_i^\varepsilon$. Integrate the two sides by parts in different ways:

$$\int \phi\, \eta^\varepsilon \cdot Dw_i^\varepsilon = \int f\phi w_i^\varepsilon - \int w_i^\varepsilon \eta^\varepsilon \cdot D\phi,$$

平均的是修正后的通量

有效系数通常不是直接对 A 求平均。微观梯度要先调整成 $e_i - D\chi_i$，使通量在单元中无散，再取平均；一维时结果是~~调和~~平均而非算术平均。

图 T4.M：一维有效系数



$\bar{a} = (\frac{1}{2} + \frac{1}{8})^{-1} = 1.6$

两个半格串联时通量相同，梯度与系数成反比。平均的是阻力 $1/a$，再取倒数。分段常系数可按弱形式理解。

$$\int \phi\, Du^\varepsilon \cdot q_i^\varepsilon = -\int u^\varepsilon q_i^\varepsilon \cdot D\phi.$$

Every product in these two right sides has a strong L^2 factor and a weak L^2 factor. The limits give $\eta_i = Du \cdot \overline{A}e_i$. By symmetry this says $\eta = \overline{A}Du$, proving (HOM4). Effective ellipticity gives uniqueness, so every subsequence has the same limit; the entire family converges. □

The expansion need not satisfy the original boundary condition termwise; boundary layers may be necessary for a quantitative error estimate. We have proved weak H^1 convergence, not strong convergence of the gradients or an $O(\varepsilon)$ global error. For nonsymmetric coefficients the source's formal cell calculation still applies with consistent index orientation; the rigorous testing argument uses adjoint correctors. That extension is not asserted here.

两个弱收敛量不能直接相乘

$A(x/\varepsilon)$ 与 Du^ε 都会振荡，不能分别取弱极限再相乘。振荡测试把乘积改写为一个强收敛因子和一个弱收敛因子的配对，才识别出有效通量。

4.6 Power Series

4.6.1 Noncharacteristic surfaces and determination of the full jet

For a quasilinear equation of order k, write

$$\sum_{|\gamma|=k} a_\gamma(D^{k-1}u, \dots, u, x) D^\gamma u + a_0(D^{k-1}u, \dots, u, x) = 0.$$

Its principal polynomial is $P(x,\zeta) = \sum_{|\gamma|=k} a_\gamma \zeta^\gamma$, with coefficients evaluated on the chosen lower-order jet. Cauchy data on a hypersurface Γ prescribe $u, \partial_\nu u, \dots, \partial_\nu^{k-1} u$. Here ∂_ν^j is the contraction of $D^j u$ with $\nu^{\otimes j}$ at the point, as in the source; iterating a variable normal field would also create lower-order curvature terms.

Theorem 4.17 (Source Theorem 1)
If $P(x,\nu) \neq 0$ along the specified data, then the PDE and the Cauchy data uniquely determine every derivative of any smooth solution along Γ. This is a determination statement, not an existence theorem.

Expanded proof.

On $\Gamma = \{x_n = 0\}$ the derivatives of the Cauchy functions in $x' = (x_1, \dots, x_{n-1})$ determine all derivatives of total order at most k, except $\partial_n^k u$. Its coefficient in the PDE is $a_{ke_n} = P(x, e_n) \neq 0$. Solving the PDE gives this missing derivative. Its tangential derivatives then give every derivative with exactly k normal differentiations. Differentiate the solved PDE once in x_n to find $\partial_n^{k+1} u$; all right-side normal orders are now at most k. Repeat, inducting on normal order and allowing arbitrary tangential differentiation at each stage.

For a curved surface use coordinates $y = \Psi(x)$ with $\Gamma = \{y_n = 0\}$. The chain rule shows that the coefficient of $\partial_{y_n}^k v$ in $u = v \circ \Psi$ is

$$\sum_{|\gamma|=k} a_\gamma (D\Psi_n)^\gamma .$$

On Γ, $D\Psi_n$ is a nonzero multiple of ν, so homogeneity and the noncharacteristic condition make this coefficient nonzero. The prescribed normal derivatives convert to y_n derivatives by a triangular system: at each order its leading coefficient is a nonzero power of $\partial_\nu \Psi_n$, while lower-order terms and tangential derivatives are already known. Apply the flat argument and transform the resulting jets back. □

For a surface $\{\rho = 0\}$, use $P(x, D\rho) \neq 0$. For the wave equation the principal polynomial is $\tau^2 - |\xi|^2$, giving characteristic light cones. For the heat equation, considered as a total-order-two PDE, the principal polynomial is $-|\xi|^2$: the time-zero surface has $(\xi, \tau) = (0, 1)$ and is characteristic. This convention does not negate forward parabolic well-posedness.

知道所有导数不等于知道函数

光滑函数可在一点具有全零 Taylor 系数而在邻域不为零，例如 e^{-1/x^2} 的零延拓。要由 jet 恢复解，需要额外的解析性或其他唯一性理论。

4.6.2 Analytic functions and coefficientwise majorants

An analytic function near zero is represented there by a convergent power series $f(x) = \sum_\gamma f_\gamma x^\gamma, f_\gamma = D^\gamma f(0)/\gamma!$. A formal series $F = \sum F_\gamma x^\gamma$ majorizes f if $F_\gamma \geq |f_\gamma|$ for every γ; write $F \gg f$.

Lemma 4.18 (Source majorant lemma, with the operations made explicit)
If F has nonnegative coefficients and converges near zero, then $F \gg f$ implies absolute convergence of f in a smaller neighborhood. Every analytic series has a majorant $C/(1 - (x_1 + \cdots + x_d)/r)$ after shrinking r. Majorization is preserved by addition, multiplication, partial differentiation, and composition with zero-constant-term series.

Expanded proof.

Coefficientwise comparison gives $\sum |f_\gamma x^\gamma| \leq F(|x_1|, \ldots, |x_d|)$. Analytic convergence on a smaller polydisc implies $|f_\gamma| \leq Cr^{-|\gamma|}$. But the coefficient of x^γ in the displayed rational series is $Cr^{-|\gamma|}|\gamma|!/\gamma! \geq Cr^{-|\gamma|}$. The multinomial expansion therefore proves the comparison. For products apply the triangle inequality to the finite coefficient convolution; for derivatives multiply each coefficient by its nonnegative integer exponent. Composition is a sum of such products. The zero constant term ensures that each fixed total-degree coefficient receives only finitely many composition contributions. These facts apply componentwise to vector and matrix series. □

4.6.3 Cauchy–Kovalevskaya: recurrence and actual convergence

After flattening an analytic noncharacteristic surface and subtracting analytic extensions of the Cauchy data, a quasilinear equation reduces to a first-order system for the derivatives of order below k. Write the normal coordinate as t and the $d = n - 1$ tangential coordinates as x':

$$U_t = \sum_{j=1}^{d} B_j(U, x')U_{x_j} + c(U, x'), \qquad U(x', 0) = 0. \tag{CK4}$$

Here U has m components. If coefficients originally depend on t, add a component $T_t = 1, T(x', 0) = 0$, and replace explicit t by T; this reduces to the displayed time-independent coefficient form.

Theorem 4.19 (Source Theorem 2)
For analytic B_j, c near $(0, 0)$, (CK4) has a unique analytic solution in a neighborhood of zero.

Expanded proof (the explicit majorant, fully verified).

1. Recursive formal solution. Write $U = \sum_{\ell \geq 1} t^\ell U_\ell(x')$. Equating the coefficient of t^ℓ gives

$$(\ell + 1)U_{\ell+1} = [t^\ell]\left\{\sum_j B_j(U, x')U_{x_j} + c(U, x')\right\}.$$

The right side only uses $U_1, \dots, U_\ell$ (and their tangential derivatives); $U_1 = c(0, x')$. Equivalently, each joint Taylor coefficient is determined recursively by smaller normal order. Composition is legitimate as a formal joint series since $U(0, 0) = 0$. No negative combinatorial coefficients occur in the operations on the right. This proves formal uniqueness, but not yet convergence.

2. Dominate the coefficient functions. Choose $r, C > 0$ so that every entry of every B_j and c is majorized by

$$F(Z, x') = \frac{Cr}{r - \sum_{j=1}^{d} x_j - \sum_{\ell=1}^{m} Z_\ell}.$$

The preceding lemma supplies such a common choice for the finite collection of analytic entries. The comparison system has all its components equal to a scalar v. With $s = \sum_j x_j$, its equation is

$$v_t = \frac{Cr}{r - s - mv}(md\, v_s + 1), \qquad v(s, 0) = 0. \tag{MAJ4}$$

控制函数值不等于控制系数

majorant 比较的是每个 Taylor 系数的绝对值。它之所以能控制递推解，是因为求导、乘积和复合只通过非负组合数来组合这些系数。

Each of the d matrix rows sums m identical derivatives, explaining the factor md.

3. Solve and check the majorant. Let $N = m(d+1)$ and set

$$v(s,t) = \frac{r - s - \sqrt{(r-s)^2 - 2NCr\,t}}{N}.$$

Use the square-root branch equal to $r - s$ near $(0,0)$. It is analytic near zero, is zero at $t = 0$, and obeys

$$(r-s)v - \frac{N}{2}v^2 = Cr\,t.$$

Implicit differentiation gives $v_t = Cr/(r-s-Nv)$ and $v_s = v/(r-s-Nv)$. Thus

$$\frac{Cr}{r-s-mv}(md\,v_s + 1) = \frac{Cr}{r-s-mv}\frac{r-s-Nv+md\,v}{r-s-Nv} = v_t,$$

because $N = md + m$. The expansion of $1 - \sqrt{1-z}$ has nonnegative coefficients; expanding $(r-s)^{-1}$ also does. Hence v is a nonnegative coefficient majorant in (x', t).

4. Compare recursively and obtain a PDE solution. At normal order zero both series vanish. Assume all coefficients up to normal order ℓ of U are dominated by those of $(v, \dots, v)$. The positive-operation rules and coefficient majorization of B_j, c give the same domination for the right side of the recurrence. Division by $\ell + 1 > 0$ preserves it. Thus the domination holds at every normal order. The convergent analytic series v bounds the absolute joint series of every component of U on a smaller polydisc. On still smaller polydiscs derivatives also converge normally. The analytic compositions are therefore genuine functions; their Taylor coefficients satisfy (CK4), so the analytic identity holds near zero. Analytic uniqueness follows from the coefficient recurrence and equality with the Taylor series. □

Returning to the scalar higher-order equation. Include every derivative $D^\gamma u$, $|\gamma| < k$, as a component. Normal derivatives of lower-order components are other components; a derivative of a top-order component either is a tangential derivative of another top-order component or, for the pure normal term, is obtained from the solved original PDE. This yields the quasilinear system above. The formal scalar jet determined by the noncharacteristic recursion satisfies these derivative identities and hence

优级数的作用是证明收敛

形式递推只确定候选解的系数。显式平方根解控制这些系数的绝对值，才把形式级数变成可求导的解析函数；数据解析本身还没有完成收敛证明。

the formal system. Its formally unique system solution is exactly the analytic series constructed above. Consequently all compatibility identities between the components hold as analytic identities. The zeroth component really is a scalar solution with the required derivatives; introducing a vector of independent unknowns did not silently discard compatibility.

For nonzero analytic initial data subtract a polynomial in t with analytic x' coefficients matching the prescribed normal jets, then incorporate its known terms into the coefficients. Analytic changes of coordinates preserve analyticity. This proves the quasilinear analytic noncharacteristic Cauchy theorem. A fully nonlinear version additionally solves its highest normal derivative by the analytic implicit function theorem and needs the corresponding nonzero derivative condition; that broader theorem is a further result, not proved by the quasilinear reduction alone.

解析存在不保证低正则范数下稳定

CK 给出非特征解析数据的局部解析解，未给出 C^0 或 Sobolev 数据到解的连续依赖。习题 2 的 Laplace Cauchy 问题正好说明存在与稳定必须分别检查。

4.7 Problems

Source: Evans §4.7, Problems 1–19. Statements are restated concisely; all solutions and domain clarifications below are editorial.

4.7.1 Separation, scaling and traveling profiles

1. Aronsson's infinity-harmonic example.

Solution. For $u = f(x) + g(y)$, the equation is $(f')^2 f'' + (g')^2 g'' = 0$. Set these terms equal to $64/81$ and $-64/81$ respectively. A resulting profile is

$$u(x, y) = |x|^{4/3} - |y|^{4/3}.$$

Off the axes, $(f')^2 f'' = 64/81$ and $(g')^2 g'' = -64/81$, so the PDE holds classically there. The function is C^1 on $\mathbb{R}^2$ but not C^2 on the axes. It is a global viscosity solution of $F(Du, D^2u) = -Du \cdot D^2u\, Du = 0$, as follows.

At $x = 0$, $y \neq 0$, no C^2 function touches u from above: the positive $|x|^{4/3}$ increment is larger than every quadratic with zero first derivative at zero. For a test ϕ touching from below, $D\phi = Du$ and $\phi_{yy} \leq u_{yy} < 0$, so $D\phi \cdot D^2\phi D\phi \leq 0$, giving $F \geq 0$. At $y = 0$, $x \neq 0$ the lower tests are absent, while an upper test has $\phi_{xx} \geq u_{xx} > 0$ and hence $F \leq 0$. At the origin every touching test has gradient zero, and $F = 0$. These are exactly the viscosity inequalities. An affine function supplies a global classical solution if that solution class is required; the nonaffine Aronsson example needs the explicit weaker interpretation. □

经典解与黏性解要分清

$|x|^{4/3} - |y|^{4/3}$ 不能在坐标轴上按经典二阶导数代入。本题用接触测试说明它的全局解意义；黏性解的一般理论留待第十章。

2. Hadamard instability of Laplace Cauchy data.

Solution. The separated ansatz $\sin(nx_1)T(x_2)$ gives $T'' = n^2 T$. The data $T(0) = 0, T'(0) = 1/n$ yield

$$u_n(x_1, x_2) = n^{-2} \sin(nx_1) \sinh(nx_2).$$

The Cauchy data tend uniformly to zero, but on any line $x_2 = h > 0$, $\sup_{x_1} |u_n| = n^{-2} \sinh(nh) \to \infty$. Thus the data-to-solution map is not continuous in these uniform data and solution norms. One must not claim pointwise divergence at every fixed x_1, since $\sin(nx_1)$ can vanish. Existence of analytic solutions does not restore this stability. □

3. A balanced bistable front.

Solution. For $f(v) = -2v^3 + 3v^2 - v$, $\int_0^1 f = 0$. The energy identity (BF4) forces speed $c = 0$ for a nonconstant front. Here $F(v) = -v^2(1 - v)^2/2$, so $(v')^2 = v^2(1 - v)^2$ and the increasing branch obeys $v' = v(1 - v)$. Hence $v(s) = (1 + e^{-(s-s_0)})^{-1}$. It has the required endpoints and vanishing derivatives. Its second derivative is $v(1 - v)(1 - 2v) = -f(v)$, verifying the ODE directly. □

4. Emden–Fowler coordinates.

Solution. Put $a = 2/(p - 1)$, $t = \log r$, and $X(t) = e^{at} v(e^t)$. Then

$$v' = r^{-a-1}(X' - aX), \quad v'' = r^{-a-2}\{X'' - (2a + 1)X' + a(a + 1)X\}.$$

Since $ap = a + 2$, the radial equation becomes the autonomous ODE

$$X'' + (n - 2 - 2a)X' + a(a - n + 2)X + X^p = 0.$$

This change treats $r > 0$; regularity at the center is an additional condition corresponding to $t \to -\infty$. □

5. Fast-diffusion scaling profile.

Solution. In the standard parabolic range $\max\{0, (n-2)/n\} < m < 1$, use $\beta = 1/(n(m-1)+2) > 0$, $\alpha = n\beta$ as in §4.2. Integrating $(W^m)' = -\beta r W$ gives

$$u(x, t) = t^{-\alpha} \left(C + \frac{(1 - m)\beta}{2m} |x|^2 t^{-2\beta} \right)^{-1/(1-m)}, \qquad C > 0.$$

It is positive and smooth at every $t > 0$. Its spatial tail is $|x|^{-2/(1-m)}$; integrability is equivalent to $2/(1-m) > n$, the stated mass-critical lower bound. Directly differentiating $W^{m-1} = C + (1-m)\beta r^2/(2m)$ verifies the profile ODE. Unlike the porous-medium profile it has no compact support. If $n = 1$, the source's written bound $(n-2)/n < m < 1$ alone also permits $m \le 0$. The above forward diffusion construction uses $m > 0$; it does not assert a decaying positive profile for those nonparabolic exponents. At $m = 0$ the positive equation is $u_t = 0$, a separate degeneracy. □

快扩散仍需正的扩散指数

扩散系数为 mu^{m-1}。$0 < m < 1$ 时在低密度处很大，产生代数尾；若 $m < 0$，最高阶符号反转，已不是同一类前向抛物扩散。

6. A boundary blowup solution.

Solution. Assume $n \ge 3$ and set $p = (n+2)/(n-2)$. For $u = A(1-r^2)^{-b}$, radial differentiation gives

$$\Delta u = 2bA(1 - r^2)^{-b-2}\{n + (2b + 2 - n)r^2\}.$$

Choose $b = (n-2)/2$ so the brace is n and $bp = b+2$. Then $-\Delta u + u^p = 0$ if

$$A = [n(n-2)]^{(n-2)/4}.$$

This is smooth and finite in the open ball but diverges at every boundary point, as required. □

7. Viscous shock profiles and the entropy direction.

Solution. For $u_t + F(u)_x - au_{xx} = 0$ put $u = v(x - ct)$. Integration yields

$$av' = F(v) - cv + b =: Q(v).$$

Separation gives $s = \int_{v(0)}^{v(s)} a/Q(z)\,dz$ wherever $Q \ne 0$. A connection from u_l to u_r forces both endpoint values of Q to vanish; otherwise v' stays away from zero near that endpoint and v cannot have the asserted limit. Subtracting the endpoint identities gives $c = (F(u_l) - F(u_r))/(u_l - u_r)$. Conversely, for $u_l > u_r$, strict convexity makes $Q < 0$ on (u_r, u_l) when b is chosen to make the endpoints zero. The zeros are simple: $Q'(u_r) < 0 < Q'(u_l)$. The integral therefore diverges to $+\infty$ as $v \downarrow u_r$ and to $-\infty$ as $v \uparrow u_l$. Its inverse is the decreasing heteroclinic profile, unique up to shift.

Normalize $v^\varepsilon(0) = (u_l + u_r)/2$. Then $v^\varepsilon(s) = v^1(s/\varepsilon)$. For $x - ct < 0$ its limit is u_l, for $x - ct > 0$ it is u_r, and on $x = ct$ it is the chosen midpoint. Boundedness gives local L^1 convergence to the Rankine–Hugoniot downward shock. The viscous connection thus selects the entropy direction. □

4.7.2 Transforms and linearization

8. Schrödinger dispersive bound.

Solution. For $g \in L^1$, the modulus of the kernel in (SC4) is $(4\pi|t|)^{-n/2}$. The triangle inequality gives $\|u(t)\|_\infty \le (4\pi|t|)^{-n/2}\|g\|_1$. The bound holds for both signs of $t \neq 0$. It is consistent with conservation of the L^2 norm, since the norms measure different features. □

9. Inverting a nonlinear heat equation.

Solution. For $y = u(x,t)$, $x = v(y,t)$ and $u_x > 0$, differentiate $u(v(y,t),t) = y$:

$$u_x = 1/v_y, \quad u_t = -v_t/v_y, \quad u_{xx} = -v_{yy}/v_y^3.$$

Substitution into $u_t = u_{xx}/u_x^2$ gives $v_t = v_{yy}$. This is local in the range of u; the inverse's spatial interval and boundary conditions may move with time. □

10. A local transformation to Liouville's equation.

Solution. Taken literally for every solution u with a smooth real $f : \mathbb{R}^3 \to \mathbb{R}$, the request is impossible: constant u would give constant $w = f(u,0,0)$, while $w_{xt} = e^w > 0$. The intended transformation must be local on a nondegenerate branch.

For $u_{xt} = 0$, write $u = A(x) + B(t)$. On $u \neq 0$, $u_x u_t > 0$, choose

$$w = f(u,u_x,u_t) = \log(2u_x u_t/u^2) = \log 2 + \log|u_x| + \log|u_t| - 2\log|u|.$$

The first two derivative terms have zero mixed derivative, while

$$(-2\log|u|)_{xt} = 2u_x u_t/u^2 = e^w.$$

This verifies the transformation on its real domain. More generally, for an additive ansatz $f = a(z) + b(p) + c(q)$, the mixed derivative equals $a''(u)u_x u_t$, which explains the logarithmic separation. □

变换的定义域不能隐去

本题对数公式只在 $u \neq 0$ 且 $u_x u_t > 0$ 的分支上成立。常数解给出一个直接检查：不存在对所有三元输入光滑、又能转换所有解的实函数。

11. Lax pairs preserve a smooth eigenvalue branch.

Solution. Differentiate $Lw = \lambda w$ and take the real Hilbert inner product with $w \neq 0$. Symmetry of L cancels the two w_t terms, leaving $\lambda' \|w\|^2 = (L_t w, w)$. If $L_t = BL - LB$, then

$$(L_t w, w) = \lambda (Bw, w) - (Bw, Lw) = 0.$$

Thus $\lambda' = 0$. For unbounded operators all these pairings require a common suitable domain and the differentiability stipulated by the exercise; formal operator symbols alone do not supply these hypotheses. □

12. The KdV commutator.

Solution. Let $D = \partial_x$, $L = -D^2 + u$, $B = -4D^3 + 6uD + 3u_x$. Using $D(av) = aDv + a_x v$ gives

$$\begin{aligned} [-4D^3, u] &= -12u_x D^2 - 12u_{xx} D - 4u_{xxx}, \\ [6uD, -D^2] &= 12u_x D^2 + 6u_{xx} D, \\ [3u_x, -D^2] &= 6u_{xx} D + 3u_{xxx}, \\ [6uD, u] &= 6uu_x. \end{aligned}$$

All derivative terms cancel, leaving $[B, L] = -u_{xxx} + 6uu_x$. Since L_t is multiplication by u_t, $(L_t - [B, L])v = (u_t + u_{xxx} - 6uu_x)v$. For a KdV solution, every eigenvalue branch meeting Problem 11's operator hypotheses is constant. This is a conditional spectral conclusion, not a domain-free proof about every possible realization of the differential expression. □

4.7.3 Asymptotics, homogenization and analytic obstructions

13. The plume's Gaussian width.

Solution. Set $A = \sigma v$ in (PL4). Then $A_t - A_{zz} + (\beta z A)_z = 0$ and $A(0) = \delta_0$. For $A = (\pi\Gamma)^{-1/2} e^{-z^2/\Gamma}$,

$$A_t/A = -\Gamma'/(2\Gamma) + z^2\Gamma'/\Gamma^2, \quad A_{zz}/A = -2/\Gamma + 4z^2/\Gamma^2.$$

Also $(\beta z A)_z/A = \beta - 2\beta z^2/\Gamma$. Both coefficients vanish precisely when $\Gamma' = 4 + 2\beta\Gamma$, with $\Gamma(0) = 0$ for the point-mass trace. Hence

$$\Gamma(t) = 4e^{2\int_0^t \beta(r)\,dr} \int_0^t e^{-2\int_0^s \beta(r)\,dr}\,ds > 0.$$

The normalized Gaussian has mass one and converges weakly to δ_0, verifying the initial condition after division by σ. □

14. A relaxation system produces logarithmic diffusion.

Solution. Write the equations as $u^\varepsilon_t + \varepsilon^{-1}u^\varepsilon_x = \varepsilon^{-2}((v^\varepsilon)^2 - (u^\varepsilon)^2)$ and the opposite equation for v^ε. Multiplying by ε^2 and using the assumed bounded remainders yields $(u^0)^2 = (v^0)^2$. Positivity forces $u^0 = v^0 = w$.

Define the bounded corrections $U^\varepsilon = (u^\varepsilon - w)/\varepsilon$ and $V^\varepsilon = (v^\varepsilon - w)/\varepsilon$; the source's remainder is only assumed to be order ε. The first equation, multiplied by ε, gives

$$(V^\varepsilon - U^\varepsilon)(u^\varepsilon + v^\varepsilon) = u^\varepsilon_x + \varepsilon u^\varepsilon_t.$$

On compact sets $w > 0$, so the denominator stays positive and $V^\varepsilon - U^\varepsilon \to w_x/(2w)$ locally uniformly. Adding the two original equations gives

$$u^\varepsilon_t + v^\varepsilon_t + \partial_x(U^\varepsilon - V^\varepsilon) = 0.$$

Pass to the limit in distributions, using the uniform derivative bounds in the hypotheses. The result is $2w_t - \partial_x(w_x/(2w)) = 0$, or

$$w_t - \tfrac{1}{4}(\log w)_{xx} = 0.$$

No separate limit of the two first-order corrections is needed. □

只能按实际余项阶数做比较

若余项仍是 ε 乘一个有界函数，就不能把固定的 u_1, v_1 自动认作唯一一阶系数。证明直接控制两者的差，再用两方程之和取极限。

15. Gaussian contour deformation.

Solution. For $q = \delta - i\lambda$, $\delta > 0$, the completed-square variable puts the contour on a translated line $e^{i\theta}\mathbb{R} + c$, where $\theta = \arg\sqrt{q}$ satisfies $|\theta| < \pi/4$. First remove the translation: close the two parallel line segments at parameters $\pm R$ by segments of bounded length. On those connectors $\operatorname{Re} z^2 \geq R^2\cos(2\theta) - CR - C$, so their integrals of e^{-z^2} tend to zero. Cauchy's theorem equates the two infinite line integrals.

Then rotate $e^{i\theta}\mathbb{R}$ onto $\mathbb{R}$. On connecting arcs of radius R within the fixed sector, $|e^{-z^2}| \leq e^{-R^2\cos(2\theta)}$; their length is $O(R)$, so they also vanish. The integral is $\sqrt{\pi}$ with the inherited orientation. This argument holds for each fixed $\delta > 0$; the limiting Fresnel branch is obtained afterward, not by claiming uniform decay on a sector boundary where the cosine is zero. □

16. Schrödinger initial recovery.

Solution. For $g \in C_c^\infty$, apply (SP4) to $\phi(y) = |x - y|^2/4$ with parameter $t > 0$. Its sole critical point is $y = x$ and Hessian is $I/2$. The factor $(4\pi t)^{n/2}e^{i\pi n/4}$ from stationary phase cancels the prefactor in (SC4), so $u(x,t) = g(x) + O(t)$, uniformly for x in compact sets. For general $g \in L^2$,

the multiplier and dominated convergence give $u(t) \to g$ strongly in L^2, as proved in §4.3. One cannot replace this by absolute-integral approximate-identity theory for a positive kernel: the Schrödinger kernel is oscillatory and is not in L^1 as a function of space. □

17. One-dimensional effective coefficient.

Solution. Set $F(x) = \int_0^x f(s)\,ds$ and $b = 1/a$. Integration of the flux equation gives

$$(u^\varepsilon)' = b(x/\varepsilon)(C_\varepsilon - F(x)), \qquad C_\varepsilon = \frac{\int_0^1 b(x/\varepsilon)F(x)\,dx}{\int_0^1 b(x/\varepsilon)\,dx}.$$

The boundary condition $\int_0^1 (u^\varepsilon)' = 0$ determines this constant. Periodic averaging gives $b(x/\varepsilon) \rightharpoonup^* \langle b\rangle$, $C_\varepsilon \to C = \int_0^1 F$. The formula is uniformly bounded in L^2, and its weak limit is $u' = \langle b\rangle(C - F)$. The zero traces and uniqueness identify the weak H_0^1 limit, with $-\overline{a}\,u'' = f$, $\overline{a} = \langle 1/a\rangle^{-1}$. For the cell problem, $a(1 - \chi')$ is constant; integrating $\chi' = 1 - \overline{a}/a$ over a period yields exactly the same harmonic mean. □

18. Primal and dual variational principles.

Solution. All averages use the unit cube, of volume one. For $\chi_\xi = \sum \xi_i \chi_i$, the cell orthogonality proves

$$\xi \cdot \overline{A}\xi = \min_{v \text{ periodic}} \int_Q (\xi + Dv) \cdot A(\xi + Dv),$$

with minimizer $v = -\chi_\xi$. For the dual problem fix η and set $\xi = \overline{A}^{-1}\eta$, $\sigma_* = A(\xi - D\chi_\xi)$. It is periodic, divergence free and has mean η. For any admissible $\sigma = \sigma_* + \tau$, $\int_Q \tau = 0$ and $\operatorname{div}\tau = 0$, so periodic integration by parts gives

$$\int_Q \tau \cdot A^{-1}\sigma_* = \int_Q \tau \cdot (\xi - D\chi_\xi) = 0.$$

Expansion of the square therefore shows that σ_* minimizes $\int_Q \sigma \cdot A^{-1}\sigma$, with minimum $\xi \cdot \overline{A}\xi = \eta \cdot \overline{A}^{-1}\eta$.

The primal competitor $v = 0$ gives $\overline{A} \le \int_Q A$. The constant dual competitor $\sigma = \eta$ gives $\overline{A}^{-1} \le \int_Q A^{-1}$. Inverting positive definite matrices reverses this order (conjugate by a square root and diagonalize), so

$$\left(\int_Q A^{-1}\right)^{-1} \le \overline{A} \le \int_Q A.$$

These are quadratic-form inequalities, not entrywise inequalities. □

19. Analytic initial data need not give a time-analytic heat solution.

Solution. The total-order-two principal polynomial of $u_t - u_{xx}$ is $-\xi^2$. At the normal to $\{t = 0\}$, $(\xi, \tau) = (0, 1)$, it vanishes. Thus this is a characteristic surface for the CK theorem.

If a solution with $u(x, 0) = 1/(1 + x^2)$ were analytic near $(0, 0)$, the PDE would give

$$\partial_t^k u(0, 0) = \partial_x^{2k}(1 + x^2)^{-1}|_{x=0} = (-1)^k (2k)!.$$

Its time Taylor series along $x = 0$ would be $\sum_{k\geq 0}(-1)^k (2k)! t^k/k!$. For every $t \neq 0$, successive term magnitudes have ratio $|t|(2k+2)(2k+1)/(k+1) \to \infty$, so terms do not even tend to zero. There is no analytic solution in a neighborhood of $(0, 0)$, and therefore none on all $\mathbb{R}^2$. The bounded forward heat solution still exists and is smooth for $t > 0$; it is time analyticity across the initial surface that fails. □

数据解析与跨初始面的解析性不同

空间初值在原点附近解析，却可能因连续两次空间求导反复产生过快的阶乘增长，使时间 Taylor 级数发散。这与正时间的热方程光滑化没有矛盾。

4.8 References

The primary source is Lawrence C. Evans, *Partial Differential Equations*, second edition, Chapter 4, printed pp. 159–236 (local PDF pp. 176–253). Section order and explicit source theorem numbers follow that edition; exposition, proof expansions, Chinese annotations, figures and exercise solutions are editorial reconstructions.

The source's §4.8 directs readers to Pinsky, Strauss, Thoe and Zachmanoglou, and Weinberger for separation; Grindrod and Nishiura for Turing instability; Linares–Ponce, Tao and Olver for dispersive equations and symmetries; Stein–Weiss, Stein, Hörmander, Rauch, Treves and Zworski for Fourier methods; Helgason, John and Lax for Radon methods; and Courant–Hilbert for hodograph and Legendre transforms. Its asymptotics discussion credits Neu, Ralston and Bensoussan–Lions–Papanicolaou; its analytic Cauchy theory refers to Folland, John and DiBenedetto. These are source reading directions, not claims of independent consultation of every listed work. The Aronsson and Hadamard exercise attributions are retained; the source credits Kurtz for Problem 14 and Kohn for Problem 18.

The Chinese chapter opening (PDF p. 129) lists the section starts at printed pp. 111, 116, 124, 136, 140, 154, 162 and 165. Selected scanned pages and the exact review range are recorded in the chapter log. A line-by-line comparison of the whole Chinese chapter has not been made.

Scope of the proofs. Spectral basis existence, periodic Poincaré, weak Hilbert compactness, Rellich and standard local hyperbolic ODE manifold theory are explicitly invoked prerequisites. Formal plume and geometric-optics expansions are kept distinct from convergence theorems. Stationary phase is proved for smooth compact amplitudes and nondegenerate critical points. The homogenization proof uses the symmetric uniformly elliptic periodic setting; it makes no global boundary-layer error claim. CK is proved in its analytic quasilinear noncharacteristic setting, not for arbitrary smooth data. Problems 1, 5 and 10 explicitly record the relevant solution classes or transformation domains instead of silently using formulas outside their domains. Numerical model checks and typesetting checks do not constitute independent mathematical review.